

dirac16complex: a complex Grassmann spinor of $\text{Pin}(4,4)$ in an arbitrary gravitational field

Lagrangian, covariant field equations, energy-momentum tensor, canonical quantization and equations of state

dirac16complex project (with Claude Opus 5.5)

September 2026

Contents

1	Scope, claims and non-claims	5
1.1	What is proved	5
1.2	What is conventional	6
1.3	What is not claimed	6
2	Counting and notation dictionary	7
2.1	Counting and complex vector spaces	7
2.2	This document, the notebook and dirac-main	8
2.3	The scalar-field PDF	9
3	The Clifford module in the split-octonion and Clifford pictures	10
3.1	The split-octonion (notebook) construction	10
3.2	Charge matrix, expression [1] and chirality	10
3.3	The Clifford picture of dirac-main and the intertwiners	12
3.4	Verification record	13
4	Definition of dirac16complex	14
4.1	The field	14
4.2	The action of $\text{Pin}(4,4)$ and the two lifts	14
4.3	Irreducibility under $\text{Pin}(4,4)$	15
4.4	Two inequivalent 8-dimensional modules under $\text{Spin}(4,4)$	15

4.5	The spinor bundle	15
5	Canonical spin connection and covariant derivative	16
5.1	Vielbein, metric and Christoffel symbols	16
5.2	The vielbein postulate and the canonical spin connection	16
5.3	Spinor connection and covariant derivatives	16
5.4	Covariant constancy of the gammas and the divergence identity	17
5.5	Local spin transformations and curvature	17
5.6	The notebook contraction and why it is wrong	18
5.7	Test geometries	18
6	Why the notebook Lagrangian $L_g[]$ is empty for a Grassmann field	19
6.1	The notebook Lagrangian, verbatim	19
6.2	Lemmas	20
6.3	Theorem	20
6.4	Machine verification	21
6.5	Remark on commuting fields	21
7	The dirac16complex Lagrangian	22
7.1	Formula	22
7.2	WolframScript form in the notebook's variable names	22
7.3	Relation to $L_g[]$, term by term	23
7.4	Hermiticity	24
7.5	Symmetries	24
7.6	The potential: polynomials only	24
7.7	Discrete maps: the chirality map and the Pin characters	25
8	Euler-Lagrange equations in an arbitrary gravitational field	26
8.1	Derivation	26
8.2	The covariant field equations	26
8.3	Component form	27
8.4	Non-triviality in an arbitrary gravitational field	27
8.5	Diagonal vielbeins	28

9	Energy-momentum tensor operator, energies, pressure and equation of state	28
9.1	Derivation from the metric variation	28
9.2	Formula	29
9.3	Properties	29
9.4	Observer decomposition	30
9.5	Kinetic and potential energy: two splits	30
9.6	Equation of state	31
9.7	Homogeneous diagonal backgrounds	31
9.8	Side-by-side with the scalar field of the PDF	32
10	Canonical quantization in 4+4 dimensions	33
10.1	Slicing and momenta	33
10.2	Constraints and Dirac brackets	33
10.3	Equal-x4 anticommutators in curved space	34
10.4	Hamiltonian and Heisenberg equations	34
10.5	Krein structure	35
10.6	Flat-space modes and the good sector	35
10.7	Fock space, Dirac sea and normal ordering	36
10.8	Unitary and Krein-unitary symmetries	36
10.9	The U(1) current and charge	38
10.10	The extra-time sector	38
10.11	Expectation values	38
11	Verification records	39
11.1	WolframScript, algebra	39
11.2	WolframScript, geometry	39
11.3	Python, algebra	40
11.4	Python, geometry	40
11.5	Python, Grassmann demonstrations	41
11.6	Agreement between the implementations	41
11.7	Files and hashes	42
12	Reproduction	43

12.1 PowerShell	43
12.2 Git Bash	44
13 Limitations	45

Abstract

This document defines `dirac16complex`, a new 16-component fermion field on an 8-dimensional manifold whose metric has signature (4,4), and derives its physics in an arbitrary gravitational field. The components $\Psi_0, \dots, \Psi_{15}$ are complex-valued and anticommuting (complex Grassmann-odd). They carry an irreducible 16-dimensional complex representation of $\text{Pin}(4,4)$ (the unique irreducible Clifford module), and under $\text{Spin}(4,4)$ the same space splits into two inequivalent irreducible 8-dimensional modules. The split-octonion gamma matrices of the author's notebook and the tensor-product gamma matrices of `dirac-main` describe one Clifford module; exact integer intertwiners, unique up to scale, connect the two pictures. The field couples to gravity only through the canonical spin connection, which is fixed by the vielbein postulate (the total covariant derivative of the vielbein vanishes). We prove that, with the canonical spin connection substituted, the notebook's Lagrangian `Lg[]` is a pure divergence for a Grassmann field, so its Euler-Lagrange equations are empty; with the notebook's own contraction its Euler-Lagrange equations are the algebraic equations $X\Psi = 0$, which force $\Psi = 0$. Either way `Lg[]` gives no wave equation, and we replace it by a Hermitian first-order Lagrangian. Its covariant field equations $\gamma^\mu D_\mu \Psi = (m + U'(S))\Psi$, $S = \bar{\Psi}\Psi$, contain the spin connection in every curved field: the connection is pure gauge exactly when the Riemann tensor vanishes, and the squared Dirac operator carries the curvature term cR with the measured constant $c = -1/4$. The energy-momentum tensor operator follows from the metric variation of the action; it is symmetric, Hermitian and conserved on shell, and its on-shell trace is $-mS + 7SU' - 8U$. In homogeneous diagonal backgrounds the energy density is $\rho = mS + U$ and the pressure is $p = SU' - U$ in all seven transverse directions. We define the kinetic and potential energies in two ways, give the equation of state $w = p/\rho$, and compare everything term by term with the scalar-field formulas $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V$, $P_\phi = \frac{1}{2}\dot{\phi}^2 - V$ of the reference PDF. Canonical quantization with respect to the time x_4 gives the equal-time anticommutator $\{\Psi, \Psi^\dagger\} = i(\sqrt{|g|} C \gamma^{x_4})^{-1} \delta^7$ and a Krein structure with fundamental symmetry $J = B = -iC\gamma^4$ of signature (8,8). The sector without momenta along the three extra times, a dimensional reduction to 4+1 dimensions, has an ordinary Fock space. Momenta along the extra times make the mode Hamiltonian non-Hermitian and non-commuting with B ; its frequency is imaginary (exponential growth) when $k_5^2 + k_6^2 + k_7^2 > m^2 + k_0^2 + \dots + k_3^2$, and at equality it is nilpotent (linear growth). Two independent exact implementations, one in WolframScript and one in Python, pass 153 of 153 checks in five reports. Their cross-comparison of 47 algebra and 84 geometry measurements finds no disagreement.

1 Scope, claims and non-claims

1.1 What is proved

Every numbered result below is backed by named checks in the five Stage-1 reports:

```
artifacts/dirac16complex/arbitrary-field/wolfram-algebra-report.json
artifacts/dirac16complex/arbitrary-field/wolfram-geometry-report.json
artifacts/dirac16complex/arbitrary-field/python-algebra-report.json
artifacts/dirac16complex/arbitrary-field/python-geometry-report.json
artifacts/dirac16complex/arbitrary-field/grassmann-demo-report.json
```

The check names are quoted in a "checks" block after each result, and Section 11 lists all of them. Numbers are copied from these reports. A statement marked "derived" follows from the checked results by the argument given, but is not itself a machine check.

- **Algebra (exact).** The Clifford relations, the charge matrix, expression [1], the chirality, the representation theory (commutants, intertwiners, invariant forms), the Krein signatures and the subgroup counts are exact integer or Gaussian-rational computations with the actual 16 by 16 matrices. Where a statement depends on parameters (masses and momenta in Section 10), the text says whether it was proved for symbolic values or checked at exact sample values.
- **Geometry (general proof, exact test points).** The statements about an arbitrary gravitational field are proved in this document by general arguments. They are then checked with exact arithmetic in three test geometries (Section 5.7): G1, a generic non-diagonal vielbein at three rational points; G2, the primordial family of Stage 2 (checked by Wolfram at three exact points and by Python at a fully symbolic point); and G3, homogeneous diagonal (Bianchi-I type) frames at three times.
- **Grassmann statements** are checked in genuine Grassmann algebras with up to 1440 odd generators (grassmann-demo-report.json), and the Euler-Lagrange checks of the Wolfram geometry verifier also use a genuine Grassmann algebra.

1.2 What is conventional

The following are choices, not results:

- counting from 0;
- the tangent metric $\eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$, with x_4 as the evolution time;
- the split-octonion basis of the notebook as the primary basis;
- the Dirac adjoint $\bar{\Psi} = \Psi^\dagger C$ with $C = \sigma_{16}$;
- the symmetrized form of the kinetic term and the default potential $U(S) = \frac{\lambda}{2} S^2$;
- the sign of $T_{\mu\nu}$, fixed so that T_{44} is the energy density;
- the two definitions of kinetic and potential energy in Section 9.5;
- units with $\hbar = 1$.

1.3 What is not claimed

1. No observational claim is made. The signature (4,4) is not the signature of observed spacetime, and nothing here is fitted to data.
2. The dark-energy and dark-matter questions are not answered in this document. Section 9 supplies the exact energy density, pressures and equation of state that such a study needs; the numerical investigation is separate work.
3. Perturbative stability of the classical field equations is not studied.

4. The quantization is canonical and formal. The state space carries an indefinite (Krein) metric (Section 10.5). Momenta along the extra times make the mode Hamiltonian non-Hermitian, with imaginary frequencies (exponential growth) when $k_5^2 + k_6^2 + k_7^2 > m^2 + k_0^2 + \dots + k_3^2$ and linear growth at equality (Section 10.10). No interacting Fock space or renormalization is constructed.
5. The homogeneous-sector formulas of Section 9.7 treat the bilinears as classical (mean-field) quantities.
6. The pairing of the masses $\pm m$ by the chirality map (Section 7.7) is a structural property of the equations, not a claim that universes of masses $\pm M$ are created in pairs.

2 Counting and notation dictionary

2.1 Counting and complex vector spaces

As the task prescribes, everything is counted from 0. A complex vector space is a set whose elements can be added and multiplied by complex numbers while satisfying the usual distributive rules. A basis $(e_0, \dots, e_7)$ of an 8-dimensional complex vector space is an ordered list in which every vector v has one and only one expansion $v = v_0 e_0 + v_1 e_1 + \dots + v_7 e_7$; the coordinate column of v is $(v_0, \dots, v_7)^T$. The dimension counts basis vectors, not the number of vectors: an 8-dimensional complex space contains infinitely many vectors. The spinor space $\mathbb{C}^{16}$ of `dirac16complex` is used in the same way, with the basis $(e_0, \dots, e_{15})$ and coordinate columns $(\Psi_0, \dots, \Psi_{15})^T$.

- Coordinates $x = \{x_0, x_1, \dots, x_7\}$; curved indices $\mu, \nu, \rho, \sigma, \lambda = 0, \dots, 7$; frame indices $a, b, c = 0, \dots, 7$; spinor indices $0, \dots, 15$.
- Roles of the coordinates (from the notebook): x_0 is a hidden space direction, x_1, x_2, x_3 are ordinary 3-space, x_4 is the evolution time, and x_5, x_6, x_7 are three extra times. Frame directions 0 to 3 are space-like ($\eta_{aa} = +1$) and 4 to 7 are time-like ($\eta_{aa} = -1$).
- A gamma matrix with a numeric upper index, $\gamma^0, \dots, \gamma^7$, is always the constant frame matrix. Curved gammas carry a Greek index, $\gamma^\mu = e_a^\mu \gamma^a$; with a numeric value they are written γ^{x_4} and so on. Lowered gammas $\gamma_\mu = g_{\mu\nu} \gamma^\nu$ are curved.
- Mathematica lists are 1-based. A notebook entry such as `SAB[[a,b]]` with $a, b = 1, \dots, 8$ is $S^{a-1 b-1}$ in this document.
- The exceptions to counting from 0 are labelled where they occur: the 1-based indices p, q, h of the notebook's 4 by 4 building blocks (Section 3.1); `dirac-main`'s labels $\gamma_k^\pm$ and $\hat{\gamma}_k^\pm$ with $k = 1, \dots, 4$ (Sections 2.2 and 3.3); the 1-based Mathematica list indices in the WolframScript code blocks, such as `$\alpha$ 1` and `A1` (Sections 6.1 and 7.2); the notebook's cell numbers and In/Out labels (Section 2.2); the numbering of this document's sections, results, lemmas and lists; and the names G1, G2, G3 of the test geometries and p1, p2, p3 and q1, q2, q3 of the test points (Section 5.7), which are names, not indices.

2.2 This document, the notebook and dirac-main

The notebook is the file

Pair_Creation_of_Universes_WaveFunctionOfUniverse-4+4-Einstein-Lovelock-Nash.nb

in the repository root (read only, never modified). "Cell N " is the N -th cell of the notebook in file order (1-based), counting only cells of the styles Input, Code, Text, Section, Subsection, Subsubsection, Title, Chapter, Item, Subtitle, DisplayFormula, ItemNumbered and Program; Output, Print and Message cells are not counted and belong to the preceding counted cell. This is the enumeration of the function loadNotebookCells in wolfram/Dirac16ComplexPrimordial.wl (Stage 2). In WolframScript, from the repository root:

```
file = First[FileNames["Pair_Creation_of_Universes*.nb"]];
counted = {"Input", "Code", "Text", "Section", "Subsection", "Subsubsection",
  "Title", "Chapter", "Item", "Subtitle", "DisplayFormula", "ItemNumbered",
  "Program"};
cells = Cases[Get[file], Cell[_ , s_String, ___] /; MemberQ[counted, s], Infinity];
Length[cells]      (* 1279 *)
cells[[1064]]      (* cell 1064, the definition of Lg[] *)
```

The cited Input cells carry these In labels: 61 In[26], 73 In[36], 147 In[78], 269 In[241], 275 In[249], 279 In[253], 283 In[258], 302 In[281], 475 In[442], 529 In[491], 532 In[494], 1058 In[1024], 1064 In[1034], 1066 In[1037], 1079 In[1052] and 1096 In[1065]; cells 99 and 100 are Text cells without a label. (The count, the labels and the snippet were checked while revising this document; they are not a recorded check.) dirac-main is the reference implementation of the Clifford and split-octonion pictures (read only).

This document	Notebook (WolframScript)	dirac-main
$x_0, \dots, x_7$	X (cell 61)	$(x_0, x_1, x_2, x_3, t, y_1, y_2, y_3): t = x_4, y_i = x_{i+4}$
η_{ab}	η_{4488}	η_{ab}
frame gamma γ^a	$(T16^{\wedge}A)[a]$	γ^a (list below)
$\gamma^\mu = e_a^\mu \gamma^a$	$(T16^{\wedge}\alpha)[\mu]$	$\gamma^\mu = e_a^\mu \gamma^a$
$C = \sigma_{16}$	σ_{16}	C (tensor picture)
$S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$	$SAB[[a+1, b+1]]$	$\Sigma_{ab} = \frac{1}{4}[\gamma_a, \gamma_b]$
$\omega_\mu^a{}_b$	$\omega_{mat}[[\mu+1, a+1, b+1]]$	$\omega_\mu^a{}_b$
$\omega_{\mu ab} = \eta_{ac} \omega_\mu^c{}_b$	not formed	$\omega_{\mu ab}$
$\Omega_\mu = \frac{1}{2} \omega_{\mu ab} S^{ab}$	$(Q1/2) \omega_{mat} SAB$, see 5.6	$\frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b]$
$D_\mu \Psi = \partial_\mu \Psi + \Omega_\mu \Psi$	$D[\Psi16, X[[\mu+1]]] + \dots$	$D_\mu \psi$
Ψ , complex Grassmann	$\Psi16$, commuting	ψ , commuting
$\bar{\Psi} = \Psi^\dagger C$	$Transpose[\Psi16].\sigma_{16}$	$\bar{\psi} = \psi^T C$
$\sqrt{ g }$	$Sqrt[detgg]$	$\sqrt{ g }$
mass m	$-HM$	no separate mass; a mass-like term $\frac{1}{20}S$ inside $V(S)$ (EINSTEIN_SPINOR_44)

This document	Notebook (WolframScript)	dirac-main
$\Psi_0, \dots, \Psi_7$	$\Psi16upper$	image of P_-
$\Psi_8, \dots, \Psi_{15}$	$\Psi16lower$	image of P_+

Further notebook names used in `Lg[]` (Section 6.1): `Q1` is the notebook's book-keeping switch for the spin connection (1 switches it on); `sg` is the list of rules that turns the metric symbols into diagonal functions of (x_0, x_4) (cell 269); `g4488` is the array of metric symbols (cell 147); `detgg` is `Det[g4488 /. sg]` (cell 275); `constraintVars` is the list of positivity assumptions used by `Simplify` (cell 73); `H` and `M` are the notebook's inverse length and mass. The curved gammas ($T16^{\wedge}\alpha$) (cell 475) are built from the notebook's diagonal vielbein (cells 279, 283 and 302), and `$\omega$ mat` (cells 529 to 532) from `g4488 /. sg`, so all of them belong to the notebook's own field in (x_0, x_4) .

The notebook's $\Psi16$ is the column of the 16 functions `fl6[k]`, which the notebook treats as real commuting functions of (x_0, x_4) . The `dirac-main` gammas form the ordered list $(\gamma_1^+, \gamma_2^+, \gamma_3^+, \gamma_4^+, \gamma_1^-, \gamma_2^-, \gamma_3^-, \gamma_4^-)$ of Section 3.3, with 1-based labels: γ_k^+ is the frame matrix γ^{k-1} and γ_k^- is γ^{k+3} ($k = 1, \dots, 4$), so the evolution-time generator γ^4 is γ_1^- .

2.3 The scalar-field PDF

The reference on the equation of state is the PDF file whose name begins with "Gmail - w = equation of state parameter" (repository root, read only). It treats a canonical scalar field ϕ in four dimensions with metric signature $(+, -, -, -)$ and $\mathcal{L}_\phi = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$.

PDF	This document
signature $(+, -, -, -)$, time first	signature (4,4), time x_4 ; the PDF metric is $-g$ on the coordinates x_1, x_2, x_3, x_4
$\mathcal{L}_\phi = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V$	the same function: $\mathcal{L}_\phi = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V$
$T^\mu{}_\nu = \partial^\mu\phi\partial_\nu\phi - \delta^\mu{}_\nu\mathcal{L}_\phi$	$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi + g_{\mu\nu}\mathcal{L}_\phi$ (same covariant components)
$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V$	$\rho = T_{44}$
$P_\phi = \frac{1}{2}\dot{\phi}^2 - V$	$p_{(i)} = T^i{}_i$ (no sum), $i \neq 4$
$w_\phi = P_\phi/\rho_\phi$	$w = p/\rho$
$\text{KE} = \frac{1}{2}\dot{\phi}^2$, $\text{PE} = V$	KE_L , PE_L and KE_H , PE_H (Section 9.5)
$\dot{\phi} = d\phi/dt$	$\partial_4 = \partial/\partial x_4$
CPL: $w(a) = w_0 + w_a(1 - a)$	not used; a is the PDF's scale factor

In both conventions a homogeneous scalar field has $\rho = \frac{1}{2}\dot{\phi}^2 + V$ and $p = \frac{1}{2}\dot{\phi}^2 - V$. The two metrics differ by an overall sign on the 4-dimensional subspace; the covariant tensor $T_{\mu\nu}$ is the same in both, and the mixed tensor $T^\mu{}_\nu$ changes sign. (These scalar-field formulas are standard and are quoted for comparison; they are not part of the machine checks.)

Sign hazard in the PDF. The PDF writes $w(a) = w_0 + w_a(1 - a)$, so $dw/da = -w_a$. Since a grows with time, "w increases with time" means $w_a < 0$, and "w decreases towards -1 " means $w_a > 0$. The PDF's table of thawing and freezing models nevertheless labels thawing ("w starts near -1 and increases over time") with $w_a > 0$ and freezing ("w starts less negative and approaches -1 ")

with $w_a < 0$. That is the reverse of what its own formula implies. Wherever thawing or freezing is meant in this project, the formula-consistent signs are used: thawing has $w_a < 0$ and freezing has $w_a > 0$.

3 The Clifford module in the split-octonion and Clifford pictures

3.1 The split-octonion (notebook) construction

The notebook builds its gamma matrices from 4 by 4 blocks. With 1-based $p, q \in \{1, 2, 3, 4\}$ and $h \in \{1, 2, 3\}$,

$$\begin{aligned} Q_a[h]_{pq} &= \epsilon_{hpq4}, & Q_b[h]_{pq} &= \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \\ s_4[h] &= Q_a[h] - Q_b[h], & t_4[h] &= Q_a[h] + Q_b[h], \\ \tau_0 &= I_8, & \tau_h &= \begin{pmatrix} 0 & s_4[h] \\ s_4[h] & 0 \end{pmatrix}, & \tau_{7-h} &= \begin{pmatrix} 0 & t_4[h] \\ -t_4[h] & 0 \end{pmatrix}, \\ \tau_7 &= \tau_1\tau_2\tau_3\tau_4\tau_5\tau_6, & \sigma &= \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}, & \bar{\tau}_0 &= I_8, & \bar{\tau}_a &= \sigma \tau_a^T \sigma \quad (a = 1, \dots, 7), \\ \gamma^a &= \begin{pmatrix} 0 & \bar{\tau}_a \\ \tau_a & 0 \end{pmatrix} & (a = 0, \dots, 7), \end{aligned}$$

where ϵ_{hpq4} is Mathematica's `Signature[{h,p,q,4}]`. In the notebook γ^a is `(T16^A)[a]` and $\bar{\tau}_a$ is `OverBar[\tau][a]`.

Result 3.1. The γ^a are integer matrices with $\{\gamma^a, \gamma^b\} = 2\eta^{ab}I_{16}$ for all 64 ordered pairs; $(\gamma^a)^2 = +1$ for $a < 4$ and -1 for $a \geq 4$. The matrix γ^a is symmetric for $a < 4$ and antisymmetric for $a \geq 4$.

checks

```
ALG_clifford, ALG_gammaTransposeSymmetry: wolfram-algebra, python-algebra
ALG_cliffordRelations: wolfram-geometry, python-geometry
```

Result 3.2 (faithfulness). The 256 ordered monomials $\gamma^{a_1} \dots \gamma^{a_k}$ ($a_1 < \dots < a_k$) are linearly independent (rank 256), and the 128 even ones have rank 128. The gammas therefore generate the full matrix algebra, $\text{Cl}(4, 4) \cong \text{Mat}_{16}(\mathbb{R})$.

checks

```
ALG_faithful: wolfram-algebra, python-algebra
```

3.2 Charge matrix, expression [1] and chirality

The charge (Dirac-adjoint) matrix is the notebook's σ_{16} :

$$C := \sigma_{16} = \gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{pmatrix} -\sigma & 0 \\ 0 & \sigma \end{pmatrix}, \quad C^T = C, \quad C^2 = I_{16}.$$

Result 3.3. $C = \gamma^0 \gamma^1 \gamma^2 \gamma^3 = \text{diag}(-\sigma, \sigma)$, C is symmetric with $C^2 = 1$, its characteristic polynomial is $(x - 1)^8(x + 1)^8$, and its signature is (8,8).

checks

```
ALG_chargeMatrix: wolfram-algebra, python-algebra, wolfram-geometry
ALG_sigma16EqualsGamma0123, ALG_CSymmetricInvolution: python-geometry
```

Expression [1] of the task, verbatim in WolframScript:

```
\[Sigma]16.(T16^A)[#]==-Transpose[\[Sigma]16.(T16^A)[#]]&/@Range[0,7]
```

It states $(C\gamma^a)^T = -C\gamma^a$ for $a = 0, \dots, 7$.

Result 3.4 (expression [1]). Expression [1] evaluates to {True, True, True, True, True, True, True, True}. The Wolfram verifier evaluates it twice: with the package's matrices, and literally, from the 11 verbatim notebook definition cells evaluated in an isolated context (the notebook's Symbolize'd names T16^A and OverBar[τ] are reproduced in two independent ways, and both reproduce the package's η , σ , τ , $\bar{\tau}$, γ^a and σ_{16} exactly).

checks

```
ALG_expression1: wolfram-algebra, python-algebra
ALG_expression1CgammaAntisymmetric: python-geometry
```

Proof. $(C\gamma^a)^T = (\gamma^a)^T C$. For $a < 4$, γ^a is symmetric and anticommutes with three of the four factors of $C = \gamma^0 \gamma^1 \gamma^2 \gamma^3$, so $\gamma^a C = -C\gamma^a$. For $a \geq 4$, γ^a is antisymmetric and anticommutes with all four factors, so $(\gamma^a)^T C = -\gamma^a C = -C\gamma^a$. In both cases $(C\gamma^a)^T = -C\gamma^a$.

Result 3.5. With $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$: $(CS^{ab})^T = -CS^{ab}$ for all 64 pairs; $C\{\gamma^c, S^{ab}\}$ is symmetric and $C[\gamma^c, S^{ab}]$ is antisymmetric for all 512 triples; $[S^{ab}, \gamma^c] = \eta^{bc}\gamma^a - \eta^{ac}\gamma^b$. The Python algebra checker counts 476 verified identities of this kind.

checks

```
ALG_spinTransposeProperties: wolfram-algebra, python-algebra
ALG_CSabAntisymmetric, ALG_CAnticommutatorGammaSSymmetric,
ALG_CCommutatorGammaSAntisymmetric, ALG_SabGammaCommutator: python-geometry
```

Result 3.6 (chirality). $\gamma^8 := \gamma^0 \gamma^1 \dots \gamma^7 = \text{diag}(-I_8, I_8)$, $(\gamma^8)^2 = 1$, γ^8 anticommutes with every γ^a and commutes with every S^{ab} and with C . The upper components $\Psi_0, \dots, \Psi_7$ (the notebook's $\Psi_{16\text{upper}}$) have chirality -1 and the lower components $\Psi_8, \dots, \Psi_{15}$ ($\Psi_{16\text{lower}}$) have chirality $+1$. The chiral projectors are $P_{\mp} = \frac{1}{2}(1 \mp \gamma^8)$.

checks

```
ALG_chirality: wolfram-algebra, python-algebra
ALG_chiralityDiag: python-geometry
```

3.3 The Clifford picture of dirac-main and the intertwiners

dirac-main builds its gammas from the 2 by 2 matrices

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$\hat{\gamma}_k^+ = G^{\otimes(k-1)} \otimes P \otimes I_2^{\otimes(4-k)}, \quad \hat{\gamma}_k^- = G^{\otimes(k-1)} \otimes N \otimes I_2^{\otimes(4-k)} \quad (k = 1, 2, 3, 4),$$

ordered $(\hat{\gamma}_1^+, \dots, \hat{\gamma}_4^+, \hat{\gamma}_1^-, \dots, \hat{\gamma}_4^-) = (\hat{\gamma}^0, \dots, \hat{\gamma}^7)$, with charge matrix $C_{\text{dm}} = \hat{\gamma}^0 \hat{\gamma}^1 \hat{\gamma}^2 \hat{\gamma}^3$ and volume element $G \otimes G \otimes G \otimes G$. Its split-octonion picture uses the Zorn vector-matrix model: a vector $x = (x_0, \dots, x_7)$ is the Zorn matrix (a, u, v, b) with $a = x_0 + x_4$, $u_i = x_i + x_{i+4}$, $v_i = -x_i + x_{i+4}$ ($i = 1, 2, 3$) and $b = x_0 - x_4$, and

$$(a, u, v, b)(c, r, w, d) = (ac + u \cdot w, ar + du - v \times w, cv + bw + u \times r, v \cdot r + bd),$$

$$\Gamma(x) = \begin{pmatrix} 0 & L_{\bar{x}} \\ L_x & 0 \end{pmatrix}, \quad \Gamma^a = \Gamma(e_a),$$

where $L_x(y) = xy$ and $\bar{x} = (x_0, -x_1, \dots, -x_7)$.

Result 3.7 (Clifford picture). The tensor matrices satisfy the Clifford relations with the same η and equal dirac-main's committed cl44-seed generators. The intertwiner equation $\hat{\gamma}^a K = K \gamma^a$ ($a = 0, \dots, 7$) has a solution space of dimension exactly 1. Its primitive integer generator K_{clifford} (first nonzero entry positive, coprime entries) has rank 16, so it is invertible, and it satisfies $KCK^{-1} = C_{\text{dm}}$ and $K\gamma^8 K^{-1} = +G \otimes G \otimes G \otimes G$. The notebook's chirality -1 block $\Psi_0, \dots, \Psi_7$ maps onto the dirac-main basis indices $\{1, 2, 4, 7, 8, 11, 13, 14\}$ and the $+1$ block onto $\{0, 3, 5, 6, 9, 10, 12, 15\}$. K_{clifford} is not a signed permutation: its entries are $0, \pm 1$ with four nonzero entries per row.

checks

ALG_cliffordPictureIntertwiner: wolfram-algebra, python-algebra

```
K_clifford =
[ 0  0  0  0  0  0  0  0  0  1 -1  0  0  1  1  0  0 ]
[-1  1  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0 ]
[ 0  0 -1  1  0  0 -1 -1  0  0  0  0  0  0  0  0  0 ]
[ 0  0  0  0  0  0  0  0  0  0  0  1 -1  0  0 -1 -1 ]
[ 0  0  1  1  0  0 -1  1  0  0  0  0  0  0  0  0  0 ]
[ 0  0  0  0  0  0  0  0  0  0  0 -1 -1  0  0 -1  1 ]
[ 0  0  0  0  0  0  0  0 -1 -1  0  0  1 -1  0  0  0 ]
[ 1  1  0  0  1 -1  0  0  0  0  0  0  0  0  0  0  0 ]
[ 1 -1  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0 ]
[ 0  0  0  0  0  0  0  0 -1  1  0  0  1  1  0  0  0 ]
[ 0  0  0  0  0  0  0  0  0  0 -1  1  0  0 -1 -1  0 ]
[ 0  0  1 -1  0  0 -1 -1  0  0  0  0  0  0  0  0  0 ]
[ 0  0  0  0  0  0  0  0  0  0  1  1  0  0 -1  1  0 ]
[ 0  0 -1 -1  0  0 -1  1  0  0  0  0  0  0  0  0  0 ]
[-1 -1  0  0  1 -1  0  0  0  0  0  0  0  0  0  0  0 ]
[ 0  0  0  0  0  0  0  0  1  1  0  0  1 -1  0  0  0 ]
```

Result 3.8 (split-octonion picture). The Zorn product has the unit e_0 , satisfies $N(xy) = N(x)N(y)$ for the norm $N(x) = \eta(x, x)$, has conjugation reversing products, and is not associative (the associator of e_1, e_2, e_4 is $2e_7$). It has 64 nonzero structure constants and equals dirac-main's split-octonion.json multiplication tensor, and the Γ^a equal dirac-main's triality44.json octonion Clifford generators. The intertwiner equation $\gamma^a K = K\Gamma^a$ has a solution space of dimension exactly 1; its primitive generator K_{octonion} has rank 16 and is block diagonal, $K_{\text{octonion}} = \text{diag}(Q, Q)$ with the matrix Q below, $Q^T Q = 2I_8$ and $Q^T \sigma Q = 2\eta$. Hence $\tau_a = QL_{e_a}Q^{-1}$ and $\bar{\tau}_a = QL_{\bar{e}_a}Q^{-1}$ for every a . The notebook's τ_a equal the octonion multiplications L_{e_a} only for $a = 0$, and no signed permutation relates them. Since $Q/\sqrt{2}$ is orthogonal, the notebook's 8-space basis is a null (light-cone) basis of the split octonions, in which the octonion norm η appears as $\sigma/2$. Finally, the product $K_{\text{clifford}} K_{\text{octonion}}$, made primitive, equals dirac-main's canonicalCliffordIntertwiner.

checks

ALG_octonionPictureIntertwiner: wolfram-algebra, python-algebra

Q =

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix}$$

Compatibility. The three sets of matrices, γ^a (notebook), $\hat{\gamma}^a$ (tensor) and Γ^a (octonion), are one Clifford module written in three bases: $\hat{\gamma}^a = K_{\text{clifford}}\gamma^a K_{\text{clifford}}^{-1}$ and $\Gamma^a = K_{\text{octonion}}^{-1}\gamma^a K_{\text{octonion}}$. Each intertwiner is unique up to a scalar, as Schur's lemma requires for an irreducible module (Section 4.3), and the notebook-to-tensor map sends C to C_{dm} and the chirality blocks to the chirality blocks. The intertwiners change coordinates only; they do not make the octonion product associative.

3.4 Verification record

Both algebra implementations build the matrices independently: the Wolfram package wolfram/Dirac16ComplexAlgebra.wl and the Python module scripts/d16c_exact.py (exact rationals and Gaussian rationals, standard library only). Python writes the exact fixture algebra-fixture.json (η , σ , τ_a , $\bar{\tau}_a$, the gammas of all three pictures, C , the chirality, the 28 S^{ab} , B and both intertwiners), and the Wolfram verifier compares every fixture matrix with its own construction: 75 of 75 matrices agree by name, none disagree and none is unmatched (ALG_fixtureAgreement in wolfram-algebra-report.json). In the other direction, the Python checker reads the Wolfram report and requires the same K_{clifford} , K_{octonion} and chirality, the same verdict for all 20 check names both reports contain, and agreement of 47 shared measurements; all 47 agree (ALG_wolframAgreement in python-algebra-report.json).

4 Definition of dirac16complex

4.1 The field

Definition. `dirac16complex` is the field

$$\Psi = (\Psi_0, \Psi_1, \dots, \Psi_{15})^T,$$

whose components are complex-valued Grassmann-odd (anticommuting) fields on an 8-dimensional manifold M with a metric of signature (4,4):

$$\begin{aligned} \Psi_a \Psi_b &= -\Psi_b \Psi_a, & \Psi_a \Psi_b^* &= -\Psi_b^* \Psi_a, & \Psi_a^* \Psi_b^* &= -\Psi_b^* \Psi_a^*, \\ (\theta_1 \theta_2)^* &= \theta_2^* \theta_1^*. \end{aligned}$$

Complex conjugation is an involution of the Grassmann algebra that reverses the order of products. We write $\Psi^\dagger = (\Psi^*)^T$ and $\bar{\Psi} = \Psi^\dagger C$. The components transform under $\text{Pin}(4,4)$ through the Clifford module of Section 3 (Section 4.2); for the transformations whose vector image has determinant 1, $\text{Spin}(4,4)$ (Section 4.2), the module splits into the two chirality blocks (Section 4.4). The conjugation rules, including the product rule and the involution property, are checked in an explicit Grassmann algebra (`GR_conjugationRules`).

4.2 The action of $\text{Pin}(4,4)$ and the two lifts

$\text{Pin}(4,4)$ is the group generated, inside $\text{Cl}(4,4)$, by the unit vectors $u = u_a \gamma^a$ with $n(u) := \eta(u, u) = \pm 1$, so that $u^2 = n(u)$. It acts on `dirac16complex` by Clifford multiplication, $\rho(u)\Psi = u\Psi$. $\text{Spin}(4,4)$ is the subgroup of products of an even number of unit vectors. $\text{Pin}(4,4)$ is the double cover of the orthogonal group $\text{O}(4,4)$ and $\text{Spin}(4,4)$ is the double cover of $\text{SO}(4,4)$. Vectors transform by the twisted adjoint action $v \mapsto \alpha(g)vg^{-1}$, where α is the parity automorphism; a unit vector u then acts as the reflection in the hyperplane orthogonal to u , and the kernel is $\{\pm 1\}$ (standard facts, not machine-checked). The untwisted adjoint action $v \mapsto gvg^{-1}$ gives, for a unit vector, minus that reflection. Because -1_8 lies in the identity component $\text{SO}_0(4,4)$ (it is the product of rotations by π in the planes (0,1), (2,3), (4,5) and (6,7)), the untwisted action also maps $\text{Pin}(4,4)$ onto $\text{O}(4,4)$. For every tested unit vector the untwisted matrix is exactly in $\text{O}(4,4)$ (`ALG_pinLiftCharacter`). The choice of lift matters for the sign characters of Section 7.7.

"Determinant 1" always means $\det \Lambda(g) = 1$ for the vector image $\Lambda(g) \in \text{O}(4,4)$, never the determinant of the 16 by 16 spinor matrix. For a unit vector $\det \Lambda(u) = -1$ under both lifts: a reflection has determinant -1 , and in 8 dimensions $\det(-R_u) = (-1)^8 \det R_u = -1$. Hence $\det \Lambda(u_1 \cdots u_k) = (-1)^k$, and $\text{Spin}(4,4) = \Lambda^{-1}(\text{SO}(4,4))$ for either lift. As 16 by 16 matrices, by contrast, all elements of $\text{Pin}(4,4)$ have determinant 1: $\det u = \pm 1$ because $u^2 = \pm 1$, it is constant on each of the connected sets $n(u) = +1$ and $n(u) = -1$, and $\det \gamma^0 = \det \gamma^4 = 1$. So the spinor determinant does not single out $\text{Spin}(4,4)$ (derived; the determinants of the eight γ^a and of both vector images for sample unit vectors were checked numerically while revising this document, not as a recorded check).

4.3 Irreducibility under $\text{Pin}(4,4)$

Theorem 4.1. $\mathbb{C}^{16}$, with $\text{Pin}(4,4)$ acting through the gamma matrices, is an irreducible complex representation of $\text{Pin}(4,4)$.

Proof. Every monomial $\gamma^{a_1} \cdots \gamma^{a_k}$ is, up to sign, a product of unit vectors, so the linear span of $\text{Pin}(4,4)$ is the span of the 256 monomials. By Result 3.2 these are linearly independent, so they span $\text{Mat}_{16}(\mathbb{R})$, and over $\mathbb{C}$ they span $\text{Mat}_{16}(\mathbb{C})$. A subspace of $\mathbb{C}^{16}$ invariant under $\text{Pin}(4,4)$ is invariant under all of $\text{Mat}_{16}(\mathbb{C})$, hence it is 0 or $\mathbb{C}^{16}$. Equivalently (Schur's lemma), the commutant $\{X : X\gamma^a = \gamma^a X \ \forall a\}$ consists of the scalars.

Result 4.1. The commutant of the eight gammas on $\mathbb{C}^{16}$ has dimension exactly 1 and consists of the scalar matrices. The dimension is computed over $\mathbb{Q}$; the coefficient matrix of the equations is rational, so its rank, and hence the solution dimension, is the same over $\mathbb{C}$.

checks

ALG_pinIrreducibleComplex, ALG_faithful: wolfram-algebra, python-algebra

4.4 Two inequivalent 8-dimensional modules under $\text{Spin}(4,4)$

Theorem 4.2. Under $\text{Spin}(4,4)$, $\mathbb{C}^{16} = \mathbb{C}_-^8 \oplus \mathbb{C}_+^8$ (the images of P_- and P_+), and the two summands are irreducible and inequivalent.

Proof. $\text{Spin}(4,4)$ is generated by even elements, and every even element commutes with γ^8 , so the two chirality blocks are invariant. The even monomials restricted to each block span an algebra of rank $64 = \dim \text{End}(\mathbb{C}^8)$, so each block is irreducible, and each block's commutant is one-dimensional. If the two blocks were equivalent, the commutant on $\mathbb{C}^{16}$ would be $\text{Mat}_2(\mathbb{C})$, of dimension 4. It has dimension 2, spanned by P_- and P_+ , and the intertwiners between the blocks form the zero space in both directions, so the blocks are inequivalent.

Result 4.2. The commutant of the 28 spin generators S^{ab} has dimension 2 and is spanned by P_- and P_+ ; every S^{ab} and every even monomial is block diagonal; the block commutants have dimensions (1, 1); the even-algebra ranks on the blocks are (64, 64); the cross-intertwiner dimensions are (0, 0).

checks

ALG_spinDecomposition: wolfram-algebra, python-algebra

The same statements hold for the identity component, which is generated by the $\exp(\theta S^{ab})$, and for the full group, whose elements lie in the even algebra.

4.5 The spinor bundle

On a manifold with a spin structure $P_{\text{Spin}}(M)$ for the (4,4) frame bundle (assumed to exist), dirac16complex is a section, with values in the odd part of a complex Grassmann algebra, of the bundle

$$S_{\mathbb{C}} = P_{\text{Spin}}(M) \times_{\rho} (\mathbb{C} \otimes \Delta_{\mathbb{R}}), \quad \Delta_{\mathbb{R}} = \Delta_- \oplus \Delta_+,$$

where $\Delta_{\mathbb{R}} \cong \mathbb{R}^{16}$ is the real module on which the real gammas act and $\Delta_{\mp}$ are its chirality halves. The spin connection of Section 5 preserves Δ_- and Δ_+ because every Ω_{μ} is even.

5 Canonical spin connection and covariant derivative

5.1 Vielbein, metric and Christoffel symbols

The gravitational field is a vielbein $e_{\mu}^a(x)$ (row index μ) with inverse e_a^{μ} :

$$g_{\mu\nu} = e_{\mu}^a \eta_{ab} e_{\nu}^b, \quad \gamma^{\mu} = e_a^{\mu} \gamma^a, \quad \{\gamma^{\mu}, \gamma^{\nu}\} = 2g^{\mu\nu},$$

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} g_{\nu\sigma} + \partial_{\nu} g_{\mu\sigma} - \partial_{\sigma} g_{\mu\nu}).$$

The Christoffel symbols are those of the torsion-free metric-compatible (Levi-Civita) connection.

5.2 The vielbein postulate and the canonical spin connection

The canonical spin connection is defined by the vielbein postulate: the total covariant derivative of the vielbein, built with the Christoffel symbols on the curved index and the spin connection on the frame index, is zero,

$$\nabla_{\mu} e_{\nu}^a := \partial_{\mu} e_{\nu}^a - \Gamma^{\rho}_{\mu\nu} e_{\rho}^a + \omega_{\mu}^a{}_b e_{\nu}^b = 0.$$

Multiplying by the inverse vielbein solves it uniquely:

$$\omega_{\mu}^a{}_b = e_b^{\nu} (\Gamma^{\rho}_{\mu\nu} e_{\rho}^a - \partial_{\mu} e_{\nu}^a), \quad \omega_{\mu ab} := \eta_{ac} \omega_{\mu}^c{}_b = -\omega_{\mu ba}.$$

The antisymmetry in (a, b) follows from metric compatibility.

Result 5.1. In G1 (three points) and G2 all $8 \cdot 8 \cdot 8 = 512$ components of the vielbein postulate vanish together with their 4096 first derivatives, and $\omega_{\mu ab} = -\omega_{\mu ba}$. The Python checker computes ω by two routes, from the Christoffel symbols and from the anholonomy coefficients, and they agree. The number of nonzero $\omega_{\mu ab}$ is 448 at each G1 point and 24 in G2.

checks

```
GEO_vielbeinPostulate_G1, GEO_vielbeinPostulate_G2,
GEO_omegaAntisymmetry_G1, GEO_omegaAntisymmetry_G2: wolfram-geometry, python-geometry
```

5.3 Spinor connection and covariant derivatives

$$\Omega_{\mu} = \frac{1}{2} \omega_{\mu ab} S^{ab} = \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b], \quad D_{\mu} \Psi = \partial_{\mu} \Psi + \Omega_{\mu} \Psi, \quad D_{\mu} \bar{\Psi} = \partial_{\mu} \bar{\Psi} - \bar{\Psi} \Omega_{\mu}.$$

The sums run over all ordered pairs (a, b) ; with a sum over $a < b$ only, the coefficient of $[\gamma^a, \gamma^b]$ is $\frac{1}{4}$. The rule for $\bar{\Psi}$ follows from $\Psi^{\dagger} \Omega_{\mu}^{\dagger} C = -\bar{\Psi} \Omega_{\mu}$, which holds because S^{ab} is real and $(S^{ab})^T C = -C S^{ab}$ (Result 3.5).

5.4 Covariant constancy of the gammas and the divergence identity

Result 5.2. The curved gammas are covariantly constant,

$$D_\mu \gamma^\nu := \partial_\mu \gamma^\nu + \Gamma^\nu_{\mu\lambda} \gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0,$$

in all 64 pairs (μ, ν) , in G1 (three points) and G2. This identity is equivalent to the vielbein postulate, since $[\Omega_\mu, \gamma^b] = -\omega_\mu{}^b{}_c \gamma^c$ by Result 3.5.

checks

GEO_gammaCovariantConstancy_G1,
GEO_gammaCovariantConstancy_G2: wolfram-geometry, python-geometry

Result 5.3 (divergence identity). Contracting $D_\mu \gamma^\mu = 0$ and using $\partial_\mu \sqrt{|g|} = \sqrt{|g|} \Gamma^\rho_{\rho\mu}$ gives

$$\partial_\mu (\sqrt{|g|} \gamma^\mu) = \sqrt{|g|} [\gamma^\mu, \Omega_\mu],$$

which holds exactly in G1 (three points) and G2; the Wolfram check also verifies $\partial_\mu \sqrt{|g|} = \sqrt{|g|} \Gamma^\rho_{\rho\mu}$. This identity is used twice: in the proof that $\text{Lg}[]$ with the canonical spin connection is a pure divergence (Section 6) and in the derivation of the field equations (Section 8).

checks

GEO_divergenceIdentity_G1, GEO_divergenceIdentity_G2: wolfram-geometry, python-geometry

5.5 Local spin transformations and curvature

Under a local spin transformation $R(x)$,

$$\Psi \rightarrow R\Psi, \quad \Omega_\mu \rightarrow R\Omega_\mu R^{-1} - (\partial_\mu R)R^{-1}, \quad D_\mu \Psi \rightarrow R D_\mu \Psi.$$

Result 5.4. For a finite local transformation $R = \gamma(n)\gamma(m)$, $\gamma(n) := n_a \gamma^a$, built from exact rational unit vector fields $n(x)$ and $m(x)$, recomputing Ω from the rotated frame gives exactly $\Omega' = R\Omega R^{-1} - (\partial R)R^{-1}$, $\gamma'^\mu = R\gamma^\mu R^{-1}$ and an unchanged metric; for $\eta(n, n)\eta(m, m) = +1$ the Lagrangian of Section 7 is unchanged (value and first derivatives). A first-order check with exact dual numbers ($\kappa^2 = 0$) gives zero first-order change of the Lagrangian.

checks

LAG_localSpinInvariance_G1, LAG_localSpinInvariance_G2: wolfram-geometry

The curvature of the spinor connection is

$$F_{\mu\nu} = \partial_\mu \Omega_\nu - \partial_\nu \Omega_\mu + [\Omega_\mu, \Omega_\nu] = +\frac{1}{2} R_{ab\mu\nu} S^{ab},$$

$$R_{ab\mu\nu} := \eta_{ac} e_\rho{}^c R^\rho{}_{\sigma\mu\nu} e_b{}^\sigma,$$

with $R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$, Ricci tensor $R_{\sigma\nu} = R^\rho_{\sigma\rho\nu}$ and scalar curvature $R = g^{\sigma\nu} R_{\sigma\nu}$.

Result 5.5. The form with $+\frac{1}{2}$ holds at all six Wolfram points and at the Python points; the form with $-\frac{1}{2}$ and the form without η (mixed indices) fail. The frame curvature equals the Riemann tensor and the Ricci tensor is symmetric.

checks

GEO_curvature_G1, GEO_curvature_G2: wolfram-geometry, python-geometry

5.6 The notebook contraction and why it is wrong

The notebook's $\omega\text{mat}[[\mu, a, b]]$ is the mixed connection $\omega_\mu^a{}_b$, and $\text{Lg}[]$ contracts it with $\text{SAB}[[a, b]] = S^{ab}$ (both indices up):

$$\Omega_\mu^{\text{nb}} = \frac{1}{2} \omega_\mu^a{}_b S^{ab} \quad (\text{one } \eta \text{ is missing}).$$

Since $\omega_\mu^a{}_b = \eta^{aa} \omega_{\mu ab}$ (no sum), the mixed components are symmetric in (a, b) whenever $\eta^{aa} = -\eta^{bb}$, that is for every space-time (boost) pair, and the antisymmetric S^{ab} deletes them. For time-time pairs $(a, b \geq 4)$ the sign is reversed. So Ω_μ^{nb} equals the space-space part of Ω_μ minus its time-time part, and has no boost part (derived from the index placement).

Result 5.6. With the notebook contraction the gammas are not covariantly constant: $D_\mu \gamma^\nu \neq 0$ in all 64 pairs (μ, ν) at every G1 point, with 4096 nonzero matrix entries, and in 15 pairs with 288 nonzero entries in G2. The largest violation $\max |D_\mu \gamma^\nu|$ is $2/3$, $1/5$ and $9/5$ at the three G2 points and approximately 1.46822, 1.52979 and 1.68937 at the three G1 points. The symmetric part of $\omega_\mu^a{}_b$ that the contraction deletes has 256 nonzero entries at each G1 point and 12 in G2. In the primordial field the notebook contraction gives $\gamma^\mu \Omega_\mu^{\text{nb}} = \frac{3H}{2}(\gamma^0 + a'_4 \gamma^4)$ instead of the correct $3H\gamma^0$, and it breaks local spin invariance already at first order. Replacing Ω by Ω^{nb} makes all eleven connection-sensitive G2 checks fail.

checks

GEO_notebookContractionFails_G1,
GEO_notebookContractionFails_G2: wolfram-geometry, python-geometry
NEG_notebookConnectionDetected_G2: wolfram-geometry

5.7 Test geometries

G1 is a generic non-diagonal vielbein,

$$e_\mu^a = \delta_\mu^a + P_\mu^a(x), \quad P_\mu^a = \frac{1}{10}((\mu+1)x_a - (a+1)x_\mu) + \frac{1}{20}x_\mu x_a + \frac{1}{30}x_{(\mu+a) \bmod 8}^2 \quad (\mu \neq a),$$

$$P_\mu^\mu = \frac{1}{10}x_\mu^2 + \frac{1}{40}x_{(\mu+1) \bmod 8},$$

evaluated at three rational points:

```

p1 = (1/7, -2/9, 1/5, 3/11, -1/13, 2/17, -3/19, 1/23)
p2 = (-1/3, 1/4, 2/7, -1/5, 1/6, -2/11, 1/9, 3/13)
p3 = (2/9, 1/8, -1/7, 1/10, -3/14, 1/12, 2/15, -1/16)

```

At each point $\det e$ is an exact positive rational (approximately 1.49738, 1.85501 and 1.37123), the metric has inertia (4, 4, 0), $g^{44} \neq 0$, and the scalar curvature is approximately -2.42693 , 13.2081 and 36.6612 . The exact rationals are in the reports, and the two implementations agree on them exactly (Section 11.6).

checks

```

GEO_frameNondegenerate_G1: wolfram-geometry, python-geometry
GEO_sqrtgSquaredEqualsDetg_G1: python-geometry

```

G2 is the primordial field of Stage 2 with an arbitrary function a_4 :

$$e_\mu^a = \text{diag}(\cot z, s^{1/6} e^{a_4} (\times 3), 1, s^{1/6} e^{-a_4} (\times 3)), \quad z = 6Hx_0, \quad t = Hx_4, \quad s = \sin z.$$

Wolfram evaluates it at three exact points; Python keeps $\sin^{1/6} z$, $\cos z$, e^{a_4} , a_4' , a_4'' , a_4''' and H as symbols. The results are $\sqrt{|g|} = \cos z$, $R = 6H^2(a_4'^2 - 7)$ (at the three Wolfram points $R = -2672/147$, $-1102/675$ and $-834/49$) and $\gamma^\mu \Omega_\mu = 3H\gamma^0$.

checks

```

GEO_primordialInvariants_G2, GEO_diagonalSlashFormula_G2: wolfram-geometry,
python-geometry
GEO_symbolicJetAgreement_G2, GEO_frameNondegenerate_G2: wolfram-geometry
GEO_sqrtgSquaredEqualsDetg_G2: python-geometry

```

G3 is a homogeneous diagonal frame, $ds^2 = -N^2 dx_4^2 + \sum_{i \neq 4} \eta_{ii} h_i^2 dx_i^2$, evaluated at $N = 1$, $h_i = 1 + x_4^2/(i+2)$ and $x_4 \in \{1/3, -2/5, 3/7\}$ (the points q1, q2 and q3 of the Python report). Fully symbolic diagonal frames are used for the metric variation (Section 9.1).

Field data at each point (the values of Ψ , $\Psi^\dagger$ and their first and second derivatives) are independent exact random rationals from recorded seeds: a documented 64-bit linear congruential generator in Wolfram, recorded seeds in Python. The Wolfram parameters are $m \in \{3/7, -2/5, 5/9\}$ and $\lambda \in \{5/11, 7/13, -3/8\}$ at p1, p2, p3.

6 Why the notebook Lagrangian Lg[] is empty for a Grassmann field

6.1 The notebook Lagrangian, verbatim

The task quotes the notebook's Lagrangian as expression [2]. Verbatim in WolframScript (the line breaks are inserted only for the page width; joining the lines without any separator gives the expression character for character):

```

Lg[]:=Sqrt[detgg] *( Transpose[\[CapitalPsi]16].\[Sigma]16.
Sum[FullSimplify[(T16^\[Alpha])\[Alpha]1-1]/.sg],constraintVars].
(D[ \[CapitalPsi]16,X[[\[Alpha]1]]]+(Q1/2)*Sum[\[Omega]mat[[\[Alpha]1,a,b]]*SAB[[a,
b]].\[CapitalPsi]16,{a,1,8},{b,1,8}]],{\[Alpha]1,1,Length[X]}]+
(H*M)*Transpose[\[CapitalPsi]16].\[Sigma]16.\[CapitalPsi]16)//Simplify[#,
constraintVars]&

```

This is the definition in notebook cell 1064 (In[1034]); its InputForm there reads $\text{Lg}[] := (\text{Simplify}[\#1, \text{constraintVars}] \&)[\text{Sqrt}[\text{detgg}] * (...)]$, and both forms parse to the identical held expression in WolframScript 1.14 (a one-off comparison made while writing this document, not one of the recorded checks). In the notation of this document, with $Q1 = 1$ and $m = -HM$,

$$\text{Lg} = \sqrt{|g|} \left[\Psi^T C \gamma^\mu (\partial_\mu \Psi + \Omega_\mu^{\text{nb}} \Psi) - m \Psi^T C \Psi \right],$$

where the notebook's $\Psi16$ is a real 16-component column and $(T16^\alpha)_{[\mu]}$ is the curved gamma γ^μ . The notebook treats the components $\text{f16}[k]$ as commuting functions. For a fermion they must anticommute, and then $\text{Lg}[]$ carries no dynamics at all.

6.2 Lemmas

Let Ψ be a real 16-component column of Grassmann-odd fields.

Lemma 6.1. For every matrix M , $\Psi^T M \Psi = \Psi^T M_A \Psi$ with $M_A = \frac{1}{2}(M - M^T)$. In particular $\Psi^T C \Psi = 0$, because C is symmetric.

Proof. $\Psi^T M \Psi = \sum_{ab} \Psi_a M_{ab} \Psi_b = -\sum_{ab} \Psi_b M_{ab} \Psi_a = -\Psi^T M^T \Psi$, so the symmetric part drops out.

Lemma 6.2. For an antisymmetric matrix function $A(x)$, $\Psi^T A \partial_\mu \Psi = \frac{1}{2} \partial_\mu (\Psi^T A \Psi) - \frac{1}{2} \Psi^T (\partial_\mu A) \Psi$.

Proof. $\partial_\mu \Psi^T A \Psi = -\Psi^T A^T \partial_\mu \Psi = \Psi^T A \partial_\mu \Psi$ by the anticommutation of the components and $A^T = -A$; the product rule gives the claim.

Lemma 6.3. $C \gamma^c S^{ab} = \frac{1}{2} C \{\gamma^c, S^{ab}\} + \frac{1}{2} C [\gamma^c, S^{ab}]$, where the first term is symmetric and the second antisymmetric (Result 3.5). Hence $\Psi^T C \gamma^\mu \Omega_\mu \Psi = \frac{1}{2} \Psi^T C [\gamma^\mu, \Omega_\mu] \Psi$.

6.3 Theorem

Theorem 6.1. For a real Grassmann-odd Ψ and the canonical spin connection, $\text{Lg}[]$ (which already contains the factor $\text{Sqrt}[\text{detgg}]$) is a pure divergence,

$$\text{Lg} = \partial_\mu V^\mu, \quad V^\mu = \frac{1}{2} \sqrt{|g|} \Psi^T C \gamma^\mu \Psi,$$

so its Euler-Lagrange operator vanishes identically: the field equations are $0 = 0$.

Proof. The mass term vanishes by Lemma 6.1. With $A^\mu = \sqrt{|g|} C \gamma^\mu$, which is antisymmetric by expression [1], Lemma 6.2 gives $\sqrt{|g|} \Psi^T C \gamma^\mu \partial_\mu \Psi = \frac{1}{2} \partial_\mu (\Psi^T A^\mu \Psi) - \frac{1}{2} \Psi^T C \partial_\mu (\sqrt{|g|} \gamma^\mu) \Psi$. By Lemma 6.3 and the divergence identity (Result 5.3) the connection term is $\sqrt{|g|} \Psi^T C \gamma^\mu \Omega_\mu \Psi = \frac{1}{2} \Psi^T C \partial_\mu (\sqrt{|g|} \gamma^\mu) \Psi$, which cancels the second term exactly. The Euler-Lagrange operator of a total divergence is identically zero.

With the notebook's own contraction Ω^{nb} in place of Ω the cancellation fails, but what remains has no derivatives: the Euler-Lagrange equations become the algebraic equations $X\Psi = 0$ with $X = \sqrt{|g|} C \sum_{\alpha} [\gamma^{\alpha}, \Omega_{\alpha}^{\text{nb}} - \Omega_{\alpha}]$. Either way `Lg[]` gives no wave equation for a Grassmann field.

6.4 Machine verification

Result 6.1. In a genuine Grassmann algebra (real Ψ and its jets as odd generators; 720 generators in the Python jet space):

1. $\Psi^T C \Psi$ has 0 monomials, while each $\Psi^T C \gamma^a \Psi$ has 8 monomials with coefficients ± 2 ; for a random integer matrix only the antisymmetric part survives.
2. $\Psi^T A \partial_{\mu} \Psi = \frac{1}{2} \partial_{\mu} (\Psi^T A \Psi)$ for each constant $C \gamma^a$ and a random antisymmetric matrix, with identically vanishing Euler-Lagrange expression; with the symmetric C instead, the kinetic term does not reduce and its Euler-Lagrange expression is $2(C \partial_0 \Psi)_0 \neq 0$.
3. For `Lg[]` itself in curved space (G1 at p1 and the symbolic G2): with the canonical Ω , 0 of the 16 Euler-Lagrange components are nonzero, and `Lg[]` equals $\partial_{\mu} V^{\mu}$ exactly. With the notebook contraction all 16 components are nonzero, contain no derivative, and equal $X\Psi$ with the X above.
4. The Wolfram geometry verifier repeats the computation at all six G1 and G2 points in its own Grassmann algebra: the Euler-Lagrange operator vanishes identically with the canonical Ω ; with the notebook contraction it is $X\Psi$ with X of rank 16, $\max |X|$ approximately 9.927, 12.36 and 10.72 at the G1 points and $3\sqrt{455}/32 \approx 1.99976$ at the first G2 point.

checks

```
GR_massTermVanishesReal, GR_bilinearOnlyAntisymmetricPartSurvives: grassmann-demo
ALG_grassmannLemmas: wolfram-geometry
GR_kineticTotalDerivativeReal, GR_kineticSymmetricMatrixContrast: grassmann-demo
GR_notebookLgELTrivial, GR_notebookLgPureDivergence: grassmann-demo
LAG_notebookLgGrassmannTrivial_G1, LAG_notebookLgGrassmannTrivial_G2: wolfram-geometry
```

Since X has rank 16 at every tested point, the equations $X\Psi = 0$ admit only $\Psi = 0$ there.

6.5 Remark on commuting fields

For commuting components (the notebook's actual use) Lemma 6.1 is reversed: only the symmetric part of a bilinear survives. The mass term $\Psi^T C \Psi$ then survives, and from the connection term only $\frac{1}{2} C \{\gamma^{\mu}, \Omega_{\mu}\}$ survives; it contains only the totally antisymmetric part $\omega_{[cab]}$ of the connection, which vanishes for every diagonal vielbein (for G2, $\sum_{\mu} \{\gamma^{\mu}, \Omega_{\mu}\} = 0$ for both contractions: `GEO_anticommutatorGammaOmegaVanishesDiagonal_G2`). For `Lg[]` as displayed in cell 1064, evaluated in a diagonal field with Clifford-consistent curved gammas $\gamma^{\mu} = e_a^{\mu} \gamma^a$, the gravitational term of the commuting-field Euler-Lagrange equations therefore comes from $\partial_{\mu}(\sqrt{|g|} \gamma^{\mu})$, not from the spin connection, and every Q1 term drops out, for either contraction.

The equations that the notebook actually stores differ from this. Its eLa (cell 1079, `Out[1052]`), the Euler-Lagrange equations of its evaluation form `La[]` (cell 1066) with the curved gammas `useT16`

of cell 1058, contains in the 8 rows 0, 1, 4, 5, 8, 9, 12 and 13 an additional Q1 term $\pm 2Hq \Psi_k$, one component k per row, with $q = Q_1 \sinh(a_4) a'_4 e^{-a_4}$ and $a'_4 = da_4/dt$, $t = Hx_4$ (row 0, for example, contains $-2HQ_1 \sinh(a_4) a'_4 e^{-a_4} \Psi_9$). After the factor $1/(2H)$ of cell 1096 the term is $\pm q \Psi_k$, and the equations of cell 1096 and later carry it. The notebook's equation for a_4 , the DSolve of Text cell 100 with the coefficient $(Q_1 - Q_1 e^{-2a_4}) a'_4 = 2q$, and its ProductLog solution in Text cell 99 are built from the same term. By the argument above this term does not come from the spin connection, and no Lg[] with Clifford-consistent curved gammas can produce it. The Stage-2 document DIRAC16COMPLEX_PRIMORDIAL_FIELD (Section 11) traces it to the substitution rule of cell 1058, which gives non-Clifford curved gammas $\gamma'^{x_5}, \gamma'^{x_6}, \gamma'^{x_7}$, and reproduces the stored eLa exactly with them. This is a second error of the notebook, in addition to the missing η of Section 5.6. This remark is derived from the lemmas and from the stored notebook output; in Stage 1 it is not a separate machine check.

7 The dirac16complex Lagrangian

7.1 Formula

The Lagrangian density of dirac16complex in an arbitrary gravitational field is

$$\mathcal{L} = \sqrt{|g|} \left[\frac{1}{2} (\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi) - m \bar{\Psi} \Psi - U(\bar{\Psi} \Psi) \right],$$

$$\bar{\Psi} = \Psi^\dagger C, \quad U(S) = \frac{\lambda}{2} S^2.$$

We write $S = \bar{\Psi} \Psi$, $\mathcal{L}_s = \mathcal{L}/\sqrt{|g|}$ and $M_{\text{eff}} = m + U'(S)$, and $K = \frac{1}{2} (\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi)$ for the kinetic term.

7.2 WolframScript form in the notebook's variable names

The same Lagrangian, written with the notebook's variable names (1-based lists, as in Lg[]):

```
(* dirac16complex Lagrangian density in the notebook's variable names. *)
(* Lists are 1-based as in the notebook: SAB[[a,b]] is S^(a-1,b-1). *)
\[CapitalPsi]16 = (f16[#] @@ X &) /@ Range[0, 15]; (* all eight coordinates *)
(* An arbitrary vielbein ev[mu,a] = e_mu^a(x0..x7), row mu. The notebook *)
(* builds g4488, detgg, (T16^alpha) and omegamat from its diagonal field *)
(* in (x0,x4) (cells 147, 275, 475, 529-532); here they are rebuilt. *)
eva = Array[ev[#1 - 1, #2 - 1] @@ X &, {8, 8}]; (* e_mu^a *)
eav = Inverse[eva]; (* e_a^mu, row a *)
g4488 = eva.\[Eta]4488.Transpose[eva]; (* g_(mu nu) *)
detgg = Det[g4488]; ginv = Inverse[g4488];
\[CapitalGamma]c = Table[(1/2) Sum[ginv[[r, s]] (D[g4488[[n, s]], X[[mu]]] +
D[g4488[[mu, s]], X[[n]]] - D[g4488[[mu, n]], X[[s]]]), {s, 1, 8}],
{r, 1, 8}, {mu, 1, 8}, {n, 1, 8}]; (* Gamma^r_(mu n) *)
Table[(T16\[Alpha])\[Alpha]1 - 1] =
Sum[eav[[A1, \[Alpha]1]] (T16^A)[A1 - 1], {A1, 1, 8}], {\[Alpha]1, 1, 8}];
(* mixed connection omega_mu^a_b from the vielbein postulate *)
\[Omega]mat = Table[Sum[eav[[b, n]] (Sum[\[CapitalGamma]c[[r, mu, n]] eva[[r, a]],
```

```

{r, 1, 8}] - D[eva[[n, a]], X[[mu]]], {n, 1, 8}],
{mu, 1, 8}, {a, 1, 8}, {b, 1, 8}];
(* lowered spin connection omega_{mu ab} = eta_{ac} omega_mu^c_b *)
\[Omega]low = Table[Sum\[Eta]4488[[a, c]] \[Omega]mat[[\[Alpha]1, c, b]], {c, 1, 8}],
{\[Alpha]1, 1, 8}, {a, 1, 8}, {b, 1, 8}];
\[CapitalOmega]16[[\[Alpha]1_] := (1/2) Sum\[Omega]low[[\[Alpha]1, a, b]] SAB[[a, b]],
{a, 1, 8}, {b, 1, 8}];
\[CapitalPsi]16bar = ConjugateTranspose\[CapitalPsi]16.\[Sigma]16;
D\[CapitalPsi]16[[\[Alpha]1_] := D\[CapitalPsi]16, X[[\[Alpha]1]]] +
\[CapitalOmega]16[[\[Alpha]1].\[CapitalPsi]16;
D\[CapitalPsi]16bar[[\[Alpha]1_] := D\[CapitalPsi]16bar, X[[\[Alpha]1]]] -
\[CapitalPsi]16bar.\[CapitalOmega]16[[\[Alpha]1];
S16 = \[CapitalPsi]16bar.\[CapitalPsi]16; (* S = Psibar Psi *)
U[s_] := (\[Lambda]/2) s^2; (* default potential *)
m = -H*M; (* mass, m = -H M *)
Ldirac16complex[] := Sqrt[Abs[detgg]]*(
(1/2) Sum[
\[CapitalPsi]16bar.(T16^\[Alpha])[[\[Alpha]1 - 1].D\[CapitalPsi]16[[\[Alpha]1] -
D\[CapitalPsi]16bar[[\[Alpha]1].(T16^\[Alpha])[[\[Alpha]1 - 1].\[CapitalPsi]16,
{\[Alpha]1, 1, Length[X]}] -
m S16 - U[S16])

```

The field depends on all eight coordinates, so Ψ_{16} is redefined accordingly. In the notebook, g4488 (cell 147), detgg (cell 275), the curved gammas (T_{16}^{α}) (cell 475) and ω_{mat} (cells 529 to 532) all belong to the notebook's own diagonal field in (x_0, x_4) (Section 2.2). For an arbitrary gravitational field the block therefore rebuilds all four from an arbitrary vielbein $\text{ev}[\mu, a]$ @@ X with the formulas of Sections 5.1 and 5.2; read with the notebook's own definitions instead, the same Lagrangian is the one in the notebook's field. While this document was revised, the block was executed, with the Symbolize'd names T_{16}^{α} and T_{16}^A replaced by plain symbols, for a non-diagonal test vielbein with space-space, space-time and time-time mixing. At a rational point the vielbein postulate holds, the curved gammas satisfy $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$, and $D_\mu \gamma^\nu = 0$ holds with Ω_{16} in all 64 pairs, whereas the notebook contraction violates it in 50 pairs. Ldirac16complex[] itself was not evaluated. This is a one-off check, not one of the recorded checks. Mathematica's Times and Dot treat these symbols as commuting, so this code is a transcription that keeps every $\bar{\Psi}$ factor to the left of every Ψ factor; it is not executed as a Grassmann computation. The verified implementation is wolfram/Dirac16ComplexGeometry.wl, which carries its own exact Grassmann algebra (convention.grassmann in wolfram-geometry-report.json).

7.3 Relation to Lg[], term by term

Lg[]	dirac16complex
Transpose[Ψ16]	ConjugateTranspose[Ψ16]
real commuting Ψ	complex Grassmann Ψ
Transpose[Ψ16].σ16	$\bar{\Psi} = \Psi^\dagger C$, the same $C = \sigma_{16}$
unsymmetrized $\Psi^T C \gamma^\mu D_\mu \Psi$	symmetrized $\frac{1}{2}(\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi)$
ω_{mat} , mixed ω_μ^{ab}	ω_{low} , lowered $\omega_{\mu ab} = \eta_{ac} \omega_\mu^c_b$
(Q1/2) with a switch Q1	the fixed factor $\frac{1}{2}$ of $\Omega_\mu = \frac{1}{2} \omega_{\mu ab} S^{ab}$
Sqrt[detgg]	Sqrt[Abs[detgg]]

<code>Lg[]</code>	<code>dirac16complex</code>
<code>+(H M) Transpose[Ψ16].σ16.Ψ16</code>	$-m\bar{\Psi}\Psi$ with $m = -HM$ (same sign)
<code>no self-interaction</code>	$-U(\bar{\Psi}\Psi)$, $U = \frac{\lambda}{2}S^2$

7.4 Hermiticity

No factor i is needed. C is real symmetric and $C\gamma^a$ is real antisymmetric (expression [1]), hence anti-Hermitian, and with the conjugation rule $(\theta_1\theta_2)^* = \theta_2^*\theta_1^*$ the symmetrized kinetic term is Hermitian. $S = \bar{\Psi}\Psi$ is Hermitian. Here and in Sections 9.3 and 10.9, "Hermitian" means invariant under the conjugation of the Grassmann algebra; for a bilinear $\Psi^\dagger M \Psi$ this is $M = M^\dagger$. After quantization such a bilinear is self-adjoint for the Krein form of Section 10.5; in the positive Hilbert-space representation it is self-adjoint only if in addition $[M, B] = 0$ (Section 10.11).

Result 7.1. $(C\gamma^a)^\dagger = -C\gamma^a$ and $C^\dagger = C$. At the coefficient level, with the coefficients taken between $\Psi^\dagger$ and Ψ , the Lagrangian has the form $\Psi^\dagger K_0 \Psi + \Psi^\dagger K^\mu \partial_\mu \Psi + \partial_\mu \Psi^\dagger L^\mu \Psi$ with $K_0 = K_0^\dagger$ and $L^\mu = (K^\mu)^\dagger$; for example $K^\mu = \frac{1}{2}\sqrt{|g|}C\gamma^\mu$ and $L^\mu = -\frac{1}{2}\sqrt{|g|}C\gamma^\mu$. With $\bar{\Psi}$ on the left the coefficients are $\tilde{K} = CK$, and the conditions read $(C\tilde{K}_0)^\dagger = C\tilde{K}_0$ and $C\tilde{L}^\mu = (C\tilde{K}^\mu)^\dagger$; the naive $\tilde{L}^\mu = (\tilde{K}^\mu)^\dagger$ fails, for example for $\tilde{K}^\mu = \frac{1}{2}\gamma^\mu$ with $\mu = 0, \dots, 3$. In the Grassmann algebra $\mathcal{L}^* = \mathcal{L}$ exactly, and S is Hermitian with 16 monomials. As a negative control, the unsymmetrized kinetic term $\bar{\Psi}\gamma^\mu D_\mu \Psi$ alone is not Hermitian.

checks

```
LAG_hermiticity_G1, LAG_hermiticity_G2: wolfram-geometry
GR_lagrangianHermitian, GR_scalarBilinearHermitian,
GR_unsymmetrizedKineticNotHermitian: grassmann-demo
```

7.5 Symmetries

- **Local spin transformations.** $\mathcal{L}$ is invariant under every local $R(x)$ of spinor norm $+1$, in particular under the identity component $\text{Spin}_0(4, 4)$ (derived from $R^T C R = C$ and the transformation law of Ω_μ ; tested in Result 5.4 for finite products of two unit vector fields and infinitesimally). Elements of spinor norm -1 are treated in Section 7.7.
- **Diffeomorphisms.** $\mathcal{L}$ is a scalar density by construction (not a separate check).
- **U(1).** $\Psi \rightarrow e^{i\alpha}\Psi$ leaves $\mathcal{L}$ invariant. The Noether current $J^\mu = -i\bar{\Psi}\gamma^\mu \Psi$ is Hermitian (GR_currentHermitian; QNT_currentHermiticity) and is conserved on shell (derived from the field equations of Section 8).

7.6 The potential: polynomials only

$S = \bar{\Psi}\Psi$ is even, of Grassmann degree 2. At a point the algebra is generated by 32 odd elements (Ψ_a and Ψ_a^*), so $S^{17} = 0$ and every function of S is a polynomial of degree at most 16. We require $U(0) = U'(0) = 0$: no constant term (no cosmological constant) and no second mass term. The default $U = \frac{\lambda}{2}S^2$ is a four-fermion contact interaction.

Result 7.2. $(\bar{\Psi}\Psi)^k$ has $\binom{16}{k}$ nonzero monomials with coefficients of absolute value $k!$ for $k = 1, \dots, 16$, namely 16, 120, 560, 1820, 4368, 8008, 11440, 12870, 11440, 8008, 4368, 1820, 560, 120, 16 and 1, and $(\bar{\Psi}\Psi)^{17} = 0$. The Grassmann derivatives of the quartic term are $\partial(\frac{\lambda}{2}S^2)/\partial\Psi_a^\dagger = \lambda S(C\Psi)_a$ (left derivative) and $\lambda S\bar{\Psi}_a$ (right derivative with respect to Ψ_a), and S commutes with Ψ .

checks

GR_quarticTermPolynomial: grassmann-demo
ALG_grassmannLemmas: wolfram-geometry

7.7 Discrete maps: the chirality map and the Pin characters

Chirality map. Under $\Psi \rightarrow \gamma^8\Psi$ one has $\bar{\Psi} \rightarrow \bar{\Psi}\gamma^8$ (because $(\gamma^8)^TC = C\gamma^8$), γ^8 commutes with D_μ , every kinetic matrix changes sign and S is unchanged. Hence

$$\begin{aligned}\mathcal{L}_{m,U}[\gamma^8\Psi] &= -\mathcal{L}_{-m,-U}[\Psi], \\ \mathcal{L}_{m,\lambda}[\gamma^8\Psi] &= -\mathcal{L}_{-m,-\lambda}[\Psi] \quad \text{for } U = \frac{\lambda}{2}S^2.\end{aligned}$$

The simpler form $\mathcal{L}_m \rightarrow -\mathcal{L}_{-m}$ holds only for $U = 0$. The energy-momentum tensor of Section 9 is built from the same bilinears and maps the same way, $T_{\mu\nu}^{(m,\lambda)}[\gamma^8\Psi] = -T_{\mu\nu}^{(-m,-\lambda)}[\Psi]$ (derived). If Ψ solves the field equations with (m, λ) , then $\gamma^8\Psi$ solves them with $(-m, -\lambda)$. This pairing of $\pm m$ is a structural property of the equations and not a physical claim. The measured signs are -1 for every kinetic matrix and $+1$ for the mass and potential terms.

checks

ALG_gamma8Map: wolfram-algebra, python-algebra

Pin characters. For a unit vector u with $u^2 = n(u)$, $\Psi \rightarrow u\Psi$ gives $\bar{\Psi}\Psi \rightarrow -n(u)\bar{\Psi}\Psi$. With the untwisted lift the kinetic term picks up the same factor $-n(u)$, so the free Lagrangian goes to $-n(u)\mathcal{L}$ and the free field equations are Pin(4,4)-covariant; with the twisted lift the kinetic factor is $+n(u)$, opposite to the mass factor. For a product u_1u_2 the factor is the spinor norm $n(u_1)n(u_2)$; the 16 products e_ae_b with $a < 4 \leq b$ have spinor norm -1 and change the sign of both terms. The even potential does not change sign: for a transformation with character -1 ,

$$\mathcal{L}_{m,\lambda} \rightarrow -\mathcal{L}_{m,-\lambda},$$

and the interacting field equation is mapped to the one with $\lambda \rightarrow -\lambda$. Exact covariance of the interacting theory under character -1 holds only for $U = 0$ or an odd U ; elements with character $+1$, in particular all of $\text{Spin}_0(4, 4)$, are exact symmetries. In the geometry verifier, a local $R = \gamma(n)\gamma(m)$ with $\eta(n, n)\eta(m, m) = -1$ maps $\mathcal{L}[m, \lambda]$ exactly to $-\mathcal{L}[m, -\lambda]$, and the naive $-\mathcal{L}[m, \lambda]$ fails for $\lambda \neq 0$.

checks

ALG_pinLiftCharacter: wolfram-algebra, python-algebra
LAG_localSpinInvariance_G1, LAG_localSpinInvariance_G2: wolfram-geometry

8 Euler-Lagrange equations in an arbitrary gravitational field

8.1 Derivation

Vary $\Psi^\dagger$ (left derivative) and Ψ (right derivative) independently. The terms of $\mathcal{L}$ that contain $\Psi^\dagger$ are $\frac{1}{2}\sqrt{|g|}\Psi^\dagger C\gamma^\mu D_\mu\Psi$, $-\frac{1}{2}\sqrt{|g|}(\partial_\mu\Psi^\dagger C - \Psi^\dagger C\Omega_\mu)\gamma^\mu\Psi$ and $-\sqrt{|g|}(mS + U)$. Integrating the derivative term by parts and using the divergence identity $\partial_\mu(\sqrt{|g|}\gamma^\mu) = \sqrt{|g|}[\gamma^\mu, \Omega_\mu]$,

$$\begin{aligned} E &:= \frac{\partial_L \mathcal{L}}{\partial \Psi^\dagger} - \partial_\mu \frac{\partial_L \mathcal{L}}{\partial (\partial_\mu \Psi^\dagger)} \\ &= \sqrt{|g|} C \left[\frac{1}{2} \gamma^\mu D_\mu \Psi + \frac{1}{2} \Omega_\mu \gamma^\mu \Psi - (m + U'(S)) \Psi \right] + \frac{1}{2} \partial_\mu (\sqrt{|g|} C \gamma^\mu \Psi) \\ &= \sqrt{|g|} C \left[\frac{1}{2} \gamma^\mu D_\mu \Psi + \frac{1}{2} \Omega_\mu \gamma^\mu \Psi + \frac{1}{2} \gamma^\mu \partial_\mu \Psi + \frac{1}{2} [\gamma^\mu, \Omega_\mu] \Psi - (m + U'(S)) \Psi \right] \\ &= \sqrt{|g|} C [\gamma^\mu D_\mu \Psi - (m + U'(S)) \Psi]. \end{aligned}$$

C is invertible, so E vanishes exactly when the Dirac equation below holds. The quartic term contributes $\lambda S C \Psi$ (Result 7.2). The variation with respect to Ψ gives the conjugate equation in the same way: with the right derivative,

$$\frac{\partial_R \mathcal{L}}{\partial \Psi_a} - \partial_\mu \frac{\partial_R \mathcal{L}}{\partial (\partial_\mu \Psi_a)} = -\sqrt{|g|} \left((D_\mu \bar{\Psi}) \gamma^\mu + (m + U'(S)) \bar{\Psi} \right)_a.$$

With the left derivative the overall sign is the opposite, because for an even $\mathcal{L}$ the left and right derivatives with respect to an odd variable differ by a sign.

8.2 The covariant field equations

$$\gamma^\mu D_\mu \Psi = (m + U'(\bar{\Psi}\Psi)) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -(m + U'(\bar{\Psi}\Psi)) \bar{\Psi},$$

with $U'(S) = \lambda S$ for the default potential. The second equation is the Dirac conjugate of the first: $(\gamma^\mu D_\mu \Psi)^\dagger C = -(D_\mu \bar{\Psi}) \gamma^\mu$ by expression [1] and Result 3.5.

Result 8.1. At every G1 and G2 point, with $\lambda \neq 0$, the Wolfram verifier finds that the Euler-Lagrange expression with respect to $\Psi_a^\dagger$ (left derivative) equals $\sqrt{|g|}(C(\gamma^\mu D_\mu \Psi - (m + \lambda S)\Psi))_a$, and that the one with respect to Ψ_a (right derivative, as in Section 8.1) equals $-\sqrt{|g|}((D_\mu \bar{\Psi})\gamma^\mu + (m + \lambda S)\bar{\Psi})_a$. It works in a genuine Grassmann algebra and computes each Euler-Lagrange operator in two independent ways (direct total derivative and a jet-commutator identity), which agree. The Python geometry checker evaluates the same two expressions, with the same signs, for $\lambda = 0$ only, with commuting exact numbers, which is exact here because the Lagrangian is bilinear with every $\bar{\Psi}$ factor on the left (measurement LAG_commutingProxyJustification). The Grassmann demonstration repeats both in a genuine Grassmann algebra at G1 p1 and in the symbolic G2, for $\lambda = 0$ and $\lambda \neq 0$ (GR_complexQuarticEL for the $\Psi^\dagger$ equation, GR_complexPsiEquation for the Ψ equation); it takes the Ψ_a derivative from the left and therefore finds the opposite overall sign, $+\sqrt{|g|}((D_\mu \bar{\Psi})\gamma^\mu + (m + \lambda S)\bar{\Psi})_a$. The case $\lambda \neq 0$ is thus covered by Wolfram and by the Grassmann demonstration, not by the Python geometry checker.

checks

```
LAG_eulerLagrangePsibar_G1,
LAG_eulerLagrangePsibar_G2, LAG_eulerLagrangePsi_G1,
LAG_eulerLagrangePsi_G2: wolfram-geometry, python-geometry
GR_complexLagrangianNonTrivial, GR_complexQuarticEL,
GR_complexPsiEquation: grassmann-demo
```

8.3 Component form

$$\gamma^\mu D_\mu \Psi = e_a^\mu \gamma^a \left(\partial_\mu + \frac{1}{8} \omega_{\mu bc} [\gamma^b, \gamma^c] \right) \Psi,$$

$$\omega_{\mu bc} = \eta_{bd} e_c^\nu (\Gamma^\rho_{\mu\nu} e_\rho^d - \partial_\mu e_\nu^d).$$

Written out, the 16 component equations are

$$\sum_{\mu=0}^7 \sum_{a=0}^7 \sum_{k=0}^{15} e_a^\mu (\gamma^a)_{jk} \left(\partial_\mu \Psi_k + \sum_{l=0}^{15} (\Omega_\mu)_{kl} \Psi_l \right) = (m + \lambda S) \Psi_j \quad (j = 0, \dots, 15),$$

$$S = \sum_{k,l=0}^{15} \Psi_k^* C_{kl} \Psi_l.$$

8.4 Non-triviality in an arbitrary gravitational field

Theorem 8.1. (a) The spin connection is locally pure gauge, $\Omega_\mu = -(\partial_\mu R)R^{-1}$ for some local spin transformation R , if and only if the Riemann tensor vanishes. (b) In a curved field no local spin transformation removes the spin connection from the field equations, and the squared Dirac operator contains the scalar curvature:

$$(\gamma^\mu D_\mu)^2 \Psi = g^{\mu\nu} (D_\mu D_\nu \Psi - \Gamma^\lambda_{\mu\nu} D_\lambda \Psi) + c R \Psi, \quad c = -\frac{1}{4}.$$

Proof. (a) A pure-gauge connection has $F_{\mu\nu} = 0$ (direct computation). Conversely $F_{\mu\nu} = \frac{1}{2} R_{ab\mu\nu} S^{ab}$ (Result 5.5), and the 28 matrices S^{ab} , $a < b$, are linearly independent (Result 3.2), so $F = 0$ forces the Riemann tensor to vanish; a flat connection is locally pure gauge. (b) If Ω could be gauged away the curvature would vanish, contradicting (a) in a curved field. The Lichnerowicz form follows from $\gamma^\mu \gamma^\nu = g^{\mu\nu} + \frac{1}{2} [\gamma^\mu, \gamma^\nu]$, $D_\mu \gamma^\nu = 0$ and $[D_\mu, D_\nu] \Psi = F_{\mu\nu} \Psi$; the constant c is fixed by the measurement below. Consequently, for $U = 0$, every solution satisfies $g^{\mu\nu} (D_\mu D_\nu \Psi - \Gamma^\lambda_{\mu\nu} D_\lambda \Psi) - \frac{1}{4} R \Psi = m^2 \Psi$: the curvature enters the second-order equation explicitly.

Result 8.2. $c = -1/4$ exactly at G1 p1, p2, p3 and at the three Wolfram G2 points and the symbolic Python G2 point, and $c = +1/4$ fails. In curved space $\gamma^\mu \Omega_\mu \neq 0$: it has 128 nonzero entries at G1 p1 (grassmann-demo) and equals $3H\gamma^0$ in G2, and the spin-connection term of the $\Psi^\dagger$ equation is nonzero in all 16 components at every G1 point and G2 sample.

checks

```
GEO_lichnerowicz_G1, GEO_lichnerowicz_G2: wolfram-geometry, python-geometry
GEO_lichnerowiczConstantSameG1G2: python-geometry
LAG_eulerLagrangePsibar_spinConnectionTermNonzeroComponents:
```

```

python-geometry measurements
GR_complexLagrangianNonTrivial: grassmann-demo (measurement
GR_complexLagrangianNonTrivial_gammaMuOmegaMu_nonzeroEntries_G1_p1)

```

8.5 Diagonal vielbeins

For a diagonal vielbein $e_\mu^a = \text{diag}(h_0, \dots, h_7)(x)$ the anticommutator part vanishes, $\{\gamma^\mu, \Omega_\mu\} = 0$, and

$$\gamma^\mu \Omega_\mu = \frac{1}{2\sqrt{|g|}} \partial_\mu (\sqrt{|g|} \gamma^\mu) = \frac{1}{2} \sum_{b=0}^7 \frac{1}{h_b} \partial_b \ln \left(\prod_{c \neq b} h_c \right) \gamma^b.$$

In the primordial field this is $3H\gamma^0$, independent of a_4 .

```

checks
GEO_diagonalSlashFormula_G2: wolfram-geometry, python-geometry
GEO_diagonalSlashFormula_G3,
GEO_anticommutatorGammaOmegaVanishesDiagonal_G2: python-geometry

```

9 Energy-momentum tensor operator, energies, pressure and equation of state

9.1 Derivation from the metric variation

The energy-momentum tensor is defined by $T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta \mathcal{S}}{\delta g^{\mu\nu}}$ with $\mathcal{S} = \int d^8x \mathcal{L}$. For a spinor the metric is varied through the vielbein, $\delta e_a^\mu = \frac{1}{2} e_{a\nu} \delta g^{\mu\nu}$ (the antisymmetric part of δe is an infinitesimal local frame rotation, under which $\mathcal{S}$ is invariant). The explicit e_a^μ in $\gamma^\mu = e_a^\mu \gamma^a$ contributes $\frac{1}{2} (\bar{\Psi} \gamma^a D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^a \Psi) \delta e_a^\mu$ and $\delta \sqrt{|g|} = -\frac{1}{2} \sqrt{|g|} g_{\mu\nu} \delta g^{\mu\nu}$; the variation of Ω_μ enters the symmetrized kinetic term only through the totally antisymmetric part of the connection and, by the standard argument, does not contribute to the symmetric tensor (for diagonal metrics the reduced Lagrangian contains no metric derivatives at all, Result 9.1). With $e_{a\nu} \gamma^a = \gamma_\nu$ this gives the formula below.

Result 9.1. For diagonal metrics the variation is carried out explicitly and reproduces the formula off shell and on shell: in the minisuperspace $h_\mu(x_4)$ with lapse $N = h_4$, $\rho = -\frac{N}{\sqrt{|g|}} \frac{\partial \mathcal{L}}{\partial N}$ and $T^i_i = \frac{h_i}{\sqrt{|g|}} \frac{\partial \mathcal{L}}{\partial h_i}$ (no sum), and Python repeats it for a local diagonal frame with symbolic h_μ and all $\partial_\nu h_\mu$. The reduced Lagrangian does not depend on the metric velocities. The off-diagonal components are not varied explicitly; they are tested through symmetry, Hermiticity and on-shell conservation in G1 (Limitations).

```

checks
EMT_variation_G3 (and measurement emt.signConvention): wolfram-geometry
EMT_variation: python-geometry

```

9.2 Formula

$$T_{\mu\nu} = -\frac{1}{4} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi + \bar{\Psi} \gamma_\nu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi \right] + g_{\mu\nu} \mathcal{L}_s,$$

$$\mathcal{L}_s = K - mS - U(S).$$

On shell $K = M_{\text{eff}} S$ (contract the two field equations with $\bar{\Psi}$ and Ψ), hence $\mathcal{L}_s = SU'(S) - U(S)$ on shell. The sign is fixed so that the energy density is T_{44} ; for the homogeneous zero mode this is $mS + U$ (Section 9.7).

checks

```
EMT_onshellLagrangianSUprimeMinusU_G1,
EMT_onshellLagrangianSUprimeMinusU_G2: python-geometry
```

9.3 Properties

Result 9.2.

1. **Symmetric:** $T_{\mu\nu} = T_{\nu\mu}$.
2. **Hermitian:** at the coefficient level, with the coefficients taken between $\Psi^\dagger$ and Ψ as in Result 7.1, $T_{\mu\nu} = \Psi^\dagger K_{\mu\nu} \Psi + \Psi^\dagger K_{\mu\nu}^\lambda \partial_\lambda \Psi + \partial_\lambda \Psi^\dagger L_{\mu\nu}^\lambda \Psi$ with Hermitian $K_{\mu\nu}$ and $L_{\mu\nu}^\lambda = (K_{\mu\nu}^\lambda)^\dagger$, and $T_{\mu\nu}^* = T_{\mu\nu}$ in the Grassmann algebra.
3. **Conserved on shell:** $\nabla^\mu T_{\mu\nu} = 0$ when the field equations and their first derivatives hold, solved exactly at the point for the x_4 -derivatives; as a negative control the off-shell divergence is nonzero.
4. **Trace:** off shell $T^\mu{}_\mu = 7K - 8(mS + U)$; on shell

$$T^\mu{}_\mu = -mS + 7SU'(S) - 8U(S) = -mS + 3\lambda S^2 \quad \text{for } U = \frac{\lambda}{2} S^2$$

The off-shell trace follows from $g^{\mu\nu} \gamma_\mu = \gamma^\nu$, which turns the bracket into $-K$, and $g^{\mu\nu} g_{\mu\nu} = 8$: $T^\mu{}_\mu = -K + 8\mathcal{L}_s$ (measurement `emt.trace` records the on-shell formula).

checks

```
EMT_symmetric_G1, EMT_symmetric_G2: python-geometry
EMT_symmetricHermitian_G1, EMT_symmetricHermitian_G2: wolfram-geometry
GR_emtHermitian: grassmann-demo
EMT_conservation_G1, EMT_conservation_G2: wolfram-geometry, python-geometry
EMT_onshellJetSolve_G1, EMT_onshellJetSolve_G2: python-geometry
EMT_traceOffShellIdentity_G1, EMT_traceOffShellIdentity_G2: python-geometry
EMT_trace_G1, EMT_trace_G2: wolfram-geometry, python-geometry
```

9.4 Observer decomposition

Take Gaussian normal time, $g_{44} = -1$, $g_{4i} = 0$, and the observer $u = \partial_4$. Then

$$\rho = T_{\mu\nu}u^\mu u^\nu = T_{44}, \quad p_{(i)} = T^i_i \quad (\text{no sum, } i \neq 4), \quad \bar{p} = \frac{1}{7} \sum_{i \neq 4} p_{(i)}.$$

There are seven transverse directions: four space-like ($i = 0, 1, 2, 3$) and three time-like ($i = 5, 6, 7$). The equation of state is $w = p/\rho$ when the transverse pressures are equal, and $w_{(i)} = p_{(i)}/\rho$ or $\bar{w} = \bar{p}/\rho$ otherwise.

9.5 Kinetic and potential energy: two splits

Split the kinetic term into its x_4 part and the rest, $K = K_4 + K_\perp$, with

$$K_4 = \frac{1}{2}(\bar{\Psi}\gamma^{x_4}D_4\Psi - (D_4\bar{\Psi})\gamma^{x_4}\Psi), \quad K_\perp = \frac{1}{2}\sum_{\mu \neq 4}(\bar{\Psi}\gamma^\mu D_\mu\Psi - (D_\mu\bar{\Psi})\gamma^\mu\Psi).$$

- **(A) Lagrangian split.** $\text{KE}_L := \frac{1}{2}K_4$ (half of the time-derivative part of the kinetic term) and $\text{PE}_L := \rho - \text{KE}_L$. This is the exact analogue of the scalar field's $\frac{1}{2}\dot{\phi}^2$ and V : $\rho = \text{KE}_L + \text{PE}_L$ by definition, and for a homogeneous isotropic state on shell also $p = \text{KE}_L - \text{PE}_L$ (Section 9.6).
- **(B) Hamiltonian split.** $\text{KE}_H := -K_\perp$ (gradient and transverse-connection energy) and $\text{PE}_H := mS + U$ (mass and self-interaction energy). In Gaussian normal time, $\gamma_4 = -\gamma^{x_4}$ and $g_{44} = -1$ turn the formula of Section 9.2 into $T_{44} = K_4 - \mathcal{L}_s = -K_\perp + mS + U$, so

$$\rho = T_{44} = \text{KE}_H + \text{PE}_H \quad \text{identically (off shell).}$$

This identity is derived here from the verified formula. The time derivatives do not appear in split (B): for a first-order field they are not an energy.

Energy density and Hamiltonian density (derived, not a machine check). In flat space in Cartesian coordinates $\rho = T_{44}$ is also the canonical Hamiltonian density of Section 10.4. In a curved field the two differ. In Gaussian normal time the Hamiltonian density of the symmetrized Lagrangian $\mathcal{L}$ itself (momenta for both Ψ and $\Psi^\dagger$) is

$$\mathcal{H}_{\mathcal{L}} = \sqrt{|g|}T_{44} - \frac{1}{2}\sqrt{|g|}\bar{\Psi}\{\gamma^{x_4}, \Omega_4\}\Psi,$$

which equals $\sqrt{|g|}\rho$ in diagonal Gaussian-normal frames ($h_4 = 1$), because $\Omega_4 = 0$ there. The Hamiltonian density $\mathcal{H}$ of $\mathcal{L}'$ (Section 10.4) carries in addition $-\frac{1}{2}\bar{\Psi}\partial_4(\sqrt{|g|}\gamma^{x_4})\Psi$. In the homogeneous frames of Section 9.7 this is $-\frac{1}{2}\theta\sqrt{|g|}\bar{\Psi}\gamma^4\Psi = -\frac{i}{2}\theta\sqrt{|g|}J^4$, with $\theta = \sum_{i \neq 4} H_i$ and $J^4 = -i\bar{\Psi}\gamma^4\Psi$ (Section 7.5), so that $\mathcal{H} = \sqrt{|g|}(\rho - \frac{i}{2}\theta J^4)$. The extra term is anti-Hermitian and nonzero whenever the background expands and the charge density is nonzero. It comes from the explicit x_4 -dependence of the total derivative added in $\mathcal{L}'$ (Section 10.4), not from the energy density.

9.6 Equation of state

For a homogeneous isotropic state on shell (Section 9.7):

$$\begin{aligned}\rho &= mS + U, & p &= SU' - U, \\ w &= \frac{p}{\rho} = \frac{SU' - U}{mS + U} = \frac{\text{KE}_L - \text{PE}_L}{\text{KE}_L + \text{PE}_L}, \\ \text{KE}_L &= \frac{1}{2}S(m + U'), & \text{PE}_L &= \frac{1}{2}(mS + 2U - SU'), & \text{KE}_H &= 0, & \text{PE}_H &= \rho.\end{aligned}$$

- **Free massive field** ($U = 0$). $\text{KE}_L = \text{PE}_L = \frac{1}{2}mS$, $p = 0$, $w = 0$: dust.
- **Default potential.** $\rho = mS + \frac{\lambda}{2}S^2$, $p = \frac{\lambda}{2}S^2$, $w = \frac{\lambda S}{2m + \lambda S}$, $\text{KE}_L = \frac{1}{2}S(m + \lambda S)$, $\text{PE}_L = \frac{1}{2}mS$.
- **Limits.** $w \rightarrow -1$ when $\text{KE}_L/\text{PE}_L \rightarrow 0$ and $w \rightarrow +1$ when $\text{PE}_L/\text{KE}_L \rightarrow 0$, as for the scalar field.
- $w > 1$. Unlike the potential $V \geq 0$ of the PDF's canonical field, $\text{PE}_L = \frac{1}{2}(mS + 2U - SU')$ has no fixed sign. For $\rho > 0$, $w > 1$ holds exactly when $\text{PE}_L < 0$; the sample of Result 9.3 has $\text{PE}_L = -75899/3780$ and $w = 75899/24059 \approx 3.155$.
- **Phantom criterion.** $\rho + p = 2\text{KE}_L = S(m + U')$. For $\rho > 0$ the state is phantom ($w < -1$) exactly when $\text{KE}_L < 0$, that is when $S(m + U') < 0$. Unlike $\frac{1}{2}\dot{\phi}^2$, KE_L has no fixed sign, because $S = \Psi^\dagger C \Psi$ is built from C of signature (8,8) (Result 3.3).

Consequence (derived from the verified formulas, not machine-checked). Section 9.7 gives $\partial_4 \rho + \sum_{i \neq 4} H_i(\rho + p_i) = 0$; with $\rho + p = S(m + U')$ this becomes $(m + U')(\partial_4 S + S \sum_i H_i) = 0$, so $S\sqrt{|g|}$ is constant wherever $m + U' \neq 0$: S dilutes as the inverse 7-volume. For the default potential $w = \lambda S/(2m + \lambda S)$ therefore changes as the 7-volume changes. For example, with $m > 0$, $\lambda < 0$ and $x = |\lambda|S/m$: $w = -x/(2 - x)$, the state is phantom with $\rho > 0$ for $1 < x < 2$, $w = -1$ at $x = 1$, and $w \rightarrow 0$ as $x \rightarrow 0$. Whether such states are physically admissible (energy conditions, stability, the quantum Krein structure) is not decided here.

9.7 Homogeneous diagonal backgrounds

For a diagonal Gaussian-normal frame $h_i(x_4)$, $h_4 = N = 1$, with Hubble rates $H_i = \partial_4 h_i/h_i$, and a homogeneous $\Psi(x_4)$:

$$\begin{aligned}\rho &= mS + U \quad (\text{off shell}), & p_{(i)} &= \mathcal{L}_s \quad (\text{off shell}) = SU' - U \quad (\text{on shell, all seven } i \neq 4), \\ T_{ij} &= \frac{1}{4}(H_i - H_j) \bar{\Psi} \gamma_i \gamma_j \gamma^{x_4} \Psi \quad (i \neq j, i, j \neq 4), & \gamma_i &= g_{ii} \gamma^{x_i}, \\ T_{4i} &= -\frac{1}{4}(\bar{\Psi} \gamma_i \partial_4 \Psi - \partial_4 \bar{\Psi} \gamma_i \Psi) \quad (\text{off shell}), & T_{4i} &= 0 \quad (\text{on shell}), \\ \partial_4 \rho + \sum_{i \neq 4} H_i (\rho + p_{(i)}) &= 0, \\ \partial_4 (\bar{\Psi} \gamma^i \gamma^j \gamma^4 \Psi) &= -\left(\sum_k H_k\right) \bar{\Psi} \gamma^i \gamma^j \gamma^4 \Psi \quad (\text{frame gammas, on shell}).\end{aligned}$$

The pressure is isotropic in all seven transverse directions although the h_i differ. The off-diagonal T_{4i} vanish on shell because $\{\gamma_i, \gamma^{x_4}\} = 0$. The T_{ij} are proportional to a tensor bilinear that decays like the inverse 7-volume, so zero initial data stay zero, and a diagonal metric is consistent with states for which this bilinear vanishes. In the flat-gamma notation of the Python checker the same component reads $T_{ij} = \frac{1}{4}\eta_{ii}\eta_{jj}h_ih_j(H_i - H_j)\bar{\Psi}\gamma^i\gamma^j\gamma^4\Psi$; the two forms are identical because $\gamma_i = \eta_{ii}h_i\gamma^i$ (notation conversion recorded in stage1-summary.json).

Result 9.3. The Wolfram verifier checks all of these at the three G3 times, off and on shell as indicated. The Python checker independently checks, with its own random on-shell data at three G3 times, the energy density, the seven equal pressures, KE_L , PE_L , $\rho = \text{KE}_L + \text{PE}_L$, $p = \text{KE}_L - \text{PE}_L$ and the form of T_{ij} . For T_{4i} it checks only the on-shell vanishing: its comparison runs on on-shell data, where both sides of the off-shell T_{4i} formula are zero, so the off-shell T_{4i} formula is checked by Wolfram only. The off-shell forms $\rho = mS + U$ and $p_{(i)} = \mathcal{L}_s$ are part of its metric-variation check. For generic states the T_{ij} are nonzero in 42 of 42 ordered pairs and the T_{4i} vanish on shell. A sample: at $x_4 = 1/3$ with $m = 4$, $\lambda = 7/6$ and $S = -75899/7560$ the Python report gives $\rho = 260864863/13996800$, $\text{KE}_L = 3793356121/97977600$, $\text{PE}_L = -75899/3780$ and $w = 75899/24059$.

checks

EMT_homogeneousReduction_G3: wolfram-geometry
 EMT_homogeneousReduction, EMT_homogeneousTimeSpaceVanishesOnShell_G3: python-geometry
 EMT_homogeneous_G3_q1 (measurement): python-geometry

9.8 Side-by-side with the scalar field of the PDF

Quantity	Scalar field (PDF)	dirac16complex (homogeneous, on shell)
Lagrangian	$\frac{1}{2}\dot{\phi}^2 - V(\phi)$	$\mathcal{L}_s = K - mS - U(S)$
energy density	$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V$	$\rho = mS + U$
pressure	$P_\phi = \frac{1}{2}\dot{\phi}^2 - V = \mathcal{L}_\phi$	$p = SU' - U = \mathcal{L}_s$
kinetic energy	$\frac{1}{2}\dot{\phi}^2 \geq 0$	$\text{KE}_L = \frac{1}{2}S'(m+U')$, either sign
potential energy	V	$\text{PE}_L = \frac{1}{2}(mS + 2U - SU')$
$\rho = \text{KE} + \text{PE}$, $p = \text{KE} - \text{PE}$	yes	yes (split A)
Hamiltonian split	$\frac{1}{2}\dot{\phi}^2 + V$	$\text{KE}_H = 0$, $\text{PE}_H = mS + U$
equation of state	$w_\phi = \frac{\frac{1}{2}\dot{\phi}^2 - V}{\frac{1}{2}\dot{\phi}^2 + V}$	$w = \frac{SU' - U}{mS + U}$
$w \rightarrow -1$	$\dot{\phi}^2 \ll V$	$ \text{KE}_L \ll \text{PE}_L$
$w \rightarrow +1$	$\dot{\phi}^2 \gg V$ (kination)	$ \text{PE}_L \ll \text{KE}_L$
$w > 1$	impossible for $V \geq 0$	possible: $\text{PE}_L < 0$ with $\rho > 0$ (sample of Result 9.3)
phantom $w < -1$	impossible for a canonical field	possible: $\text{KE}_L < 0$ with $\rho > 0$
equation of motion	$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0$	$\gamma^\mu D_\mu \Psi = (m + U')\Psi$
dilution	depends on V	$S \propto 1/\sqrt{ g }$ (derived, 9.6)

Exact analogy. In both cases the energy density is T_{44} , and for a homogeneous state the pressure equals the Lagrangian density. In flat space ρ is also the canonical Hamiltonian density. In the

homogeneous diagonal backgrounds $\sqrt{|g|} \rho$ is the Hamiltonian density of the symmetrized Lagrangian $\mathcal{L}$, as $a^3 \rho_\phi$ is for the scalar field, whereas the Hamiltonian density of $\mathcal{L}'$ (Section 10.4) differs from it by $-\frac{i}{2} \theta \sqrt{|g|} J^4$ in an expanding background (Section 9.5). As for the scalar field, the Lagrangian split gives $\rho = \text{KE} + \text{PE}$ and $p = \text{KE} - \text{PE}$, hence the same formula for w . **Difference.** The fermion Lagrangian is first order in time. Its time-derivative term is not a positive kinetic energy: in the Hamiltonian split it contributes nothing to ρ , and in the Lagrangian split $\text{KE}_L = \frac{1}{2} S(m + U')$ can be negative. The canonical scalar field satisfies $w \geq -1$ whenever $\rho > 0$; `dirac16complex` can cross $w = -1$ at the level of these classical (mean-field) formulas.

10 Canonical quantization in 4+4 dimensions

10.1 Slicing and momenta

We quantize with respect to x_4 on the 7-dimensional slices $x_4 = \text{const}$, with coordinates $(x_0, x_1, x_2, x_3, x_5, x_6, x_7)$. This requires $g^{44} \neq 0$ (the slices are non-characteristic). Split $\mathcal{L}$ into its x_4 part and the rest and add the total derivative $\frac{1}{2} \partial_4 (\sqrt{|g|} \bar{\Psi} \gamma^{x_4} \Psi)$:

$$\mathcal{L}' = \mathcal{L} + \frac{1}{2} \partial_4 (\sqrt{|g|} \bar{\Psi} \gamma^{x_4} \Psi) = \sqrt{|g|} \Psi^\dagger C \gamma^{x_4} \partial_4 \Psi + (\text{terms without } \partial_4 \Psi \text{ or } \partial_4 \Psi^\dagger).$$

The momentum conjugate to Ψ (right derivative) is $\Pi = \sqrt{|g|} \Psi^\dagger C \gamma^{x_4}$, and $\mathcal{L}'$ does not depend on $\partial_4 \Psi^\dagger$.

Result 10.1. In a genuine Grassmann algebra, $\Pi_a = \partial_R \mathcal{L}' / \partial (\partial_4 \Psi_a) = \sqrt{|g|} (\Psi^\dagger C \gamma^{x_4})_a$ and $\partial_L \mathcal{L}' / \partial (\partial_4 \Psi_a^\dagger) = 0$ at every G1 and G2 point.

checks

`QNT_canonicalMomentum_G1, QNT_canonicalMomentum_G2: wolfram-geometry`

10.2 Constraints and Dirac brackets

The momenta do not contain velocities, so there are primary constraints $\chi_a = \Pi_a - \sqrt{|g|} (\Psi^\dagger C \gamma^{x_4})_a \approx 0$ and $\bar{\chi}_a = \Pi_{\Psi^\dagger, a} \approx 0$. Their graded Poisson brackets form, up to sign and transposition, the matrix $\mathcal{K} = \sqrt{|g|} C \gamma^{x_4}$. It is invertible exactly when $g^{44} \neq 0$, because

$$(C \gamma^{x_4})(\gamma^{x_4} C) = (\gamma^{x_4} C)(C \gamma^{x_4}) = g^{44} I_{16}, \quad (C \gamma^{x_4})^{-1} = \frac{\gamma^{x_4} C}{g^{44}},$$

so the constraints are second class. The Dirac bracket eliminates them, and the quantization rule $\{\cdot, \cdot\}_+ = i\{\cdot, \cdot\}_D$ gives the anticommutators below. The matrix identity is verified symbolically for an arbitrary e_a^4 and at the G1 points. The graded constraint-matrix computation itself is not a separate machine check; the verifiers check the result as the matrix identity $\{\Psi, \Psi^\dagger\} = i\mathcal{K}^{-1}$ that follows from the first-order form of $\mathcal{L}'$.

checks

`QNT_curvedAnticommutatorMatrix: wolfram-algebra`

10.3 Equal-x4 anticommutators in curved space

For x and y on the same slice ($\hbar = 1$):

$$\begin{aligned}\{\Psi_a(x), \Psi_b^\dagger(y)\} &= i[(C\gamma^{x_4})^{-1}]_{ab} \frac{\delta^7(x-y)}{\sqrt{|g|}} = i \frac{[\gamma^{x_4}C]_{ab}}{g^{44}} \frac{\delta^7(x-y)}{\sqrt{|g|}}, \\ \{\Psi_a(x), \Psi_b(y)\} &= \{\Psi_a^\dagger(x), \Psi_b^\dagger(y)\} = 0.\end{aligned}$$

The matrix $i(\sqrt{|g|}C\gamma^{x_4})^{-1}$ is Hermitian. In Gaussian normal gauge ($g^{44} = -1$ and $\gamma^{x_4} = \gamma^4$, as in the primordial field G2) this becomes

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = B_{ab} \frac{\delta^7(x-y)}{\sqrt{|g|}}, \quad B := -iC\gamma^4.$$

Result 10.2. B is Hermitian, $B^2 = 1$, its eigenvalues are $+1$ and -1 with multiplicity 8 each (characteristic polynomial $(x-1)^8(x+1)^8$), $[C, B] = 0$ and $BC = -i\gamma^4$. The Gaussian-normal reduction of the anticommutator to $B/\sqrt{|g|}$ is checked in the primordial field (measurement QNT_curvedAnticommutatorMatrix.G2GaussianNormalReducesToB); the canonical-momentum checks record the general formula.

checks

```
ALG_chargeFormB: wolfram-algebra, python-algebra
QNT_curvedAnticommutatorMatrix: wolfram-algebra
QNT_canonicalMomentum_G1, QNT_canonicalMomentum_G2: wolfram-geometry
```

10.4 Hamiltonian and Heisenberg equations

The canonical Hamiltonian density is $\mathcal{H} = \Pi \partial_4 \Psi - \mathcal{L}'$. Using the split of Section 9.5 (derived here, not a separate machine check):

$$\begin{aligned}\mathcal{H} &= \sqrt{|g|} (\text{KE}_H + \text{PE}_H) \\ &\quad - \frac{1}{2} \sqrt{|g|} \bar{\Psi} \{\gamma^{x_4}, \Omega_4\} \Psi - \frac{1}{2} \bar{\Psi} \partial_4 (\sqrt{|g|} \gamma^{x_4}) \Psi.\end{aligned}$$

The last two terms come from the connection along x_4 and from the explicit x_4 -dependence of the added total derivative. In flat space in Cartesian coordinates both vanish and $\mathcal{H} = T_{44} = \rho$. In the homogeneous diagonal frames of Section 9.7 the first vanishes ($\Omega_4 = 0$) but the second does not, and $\mathcal{H} = \sqrt{|g|}(\rho - \frac{i}{2}\theta J^4)$ with $\theta = \sum_{i \neq 4} H_i$ (Section 9.5). In flat space, with $\{\Psi_a, \Psi_b^\dagger\} = B_{ab}\delta^7$ and $U = 0$, $H = \int d^7x \Psi^\dagger (mC - C\gamma^j \partial_j) \Psi$ (sum over $j \neq 4$), and the Heisenberg equation gives

$$\partial_4 \Psi = i[H, \Psi] = -iB(mC - C\gamma^j \partial_j) \Psi = -m\gamma^4 \Psi + \gamma^4 \gamma^j \partial_j \Psi,$$

which is the field equation $\gamma^\mu \partial_\mu \Psi = m\Psi$ multiplied by γ^4 (using $BC = -i\gamma^4$ and $(\gamma^4)^2 = -1$).

10.5 Krein structure

Theorem 10.1. Every $\text{Spin}(4,4)$ -invariant Hermitian form on $\mathbb{C}^{16}$ is even in chirality, whereas γ^4 is odd. Consequently every charge density of the form $\text{Herm}(cH\gamma^4)$ built from an invariant form H is indefinite, of signature (8,8), or vanishes. The canonical state space is therefore a Krein space; this is intrinsic to signature (4,4) and not a consequence of the particular choice B .

Result 10.3. The $\text{Spin}(4,4)$ -invariant bilinear forms form a 2-dimensional space spanned by CP_- and CP_+ (both symmetric and block diagonal in chirality); an invariant Hermitian form is $H = \alpha CP_- + \beta CP_+$ with real α, β ; $H\gamma^4$ is block off-diagonal, traceless and similar to its negative, with $(H\gamma^4)^2 = -\alpha\beta I_{16}$; and for every constant c the Hermitian part of $cH\gamma^4$ is traceless with square $|k|^2 I_{16}$, $k = \frac{1}{2}(\bar{c}\beta - c\alpha)$, hence has signature (8,8) unless it vanishes.

checks

ALG_invariantForms: wolfram-algebra, python-algebra

The canonical anticommutator defines the Krein form $[f, h] = f^\dagger B h$ on the one-particle space. $J = B$ is a fundamental symmetry ($J = J^\dagger = J^{-1}$), and the associated positive Hilbert product is $[f, Jh] = f^\dagger h$, the standard one. In a Hilbert-space representation the Hilbert adjoint of Ψ is $\chi := \Psi^\dagger B$, with $\{\Psi_a, \chi_b\} = \delta_{ab} \delta^7$ (Gaussian normal gauge, flat): the canonical $\Psi^\dagger = \chi B$ is the Krein adjoint of Ψ , not its Hilbert adjoint.

Result 10.4. At rest the positive- and negative-frequency spaces (the eigenspaces of $-i\gamma^4$, since $h_0 = m(-i\gamma^4)$) are 8-dimensional, B preserves them, and the B -form on each has signature (4,4). The same (4,4) holds for moving states in the good sector: at $m = 1$, $k = (1, 1, 2, 3)$, $E = 4$ for both signs of the energy, and at $m = 3$, $k = (1, 1, 1, 2)$, $E = 4$ for positive energy. The Krein form is thus indefinite even among positive-energy states.

checks

QNT_kreinSignature: wolfram-algebra, python-algebra

10.6 Flat-space modes and the good sector

In flat space the field equation takes the Schrödinger form $i\partial_4 \Psi = h\Psi$ with, for a plane wave $e^{ik_j x^j}$ ($j \in \{0, 1, 2, 3, 5, 6, 7\}$),

$$h_k = -im\gamma^4 - \gamma^4 \sum_{j \neq 4} k_j \gamma^j, \quad [h_k, B] = 2i \sum_{j=5}^7 k_j C \gamma^j,$$

$$h_k^2 = (m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2 - k_5^2 - k_6^2 - k_7^2) I_{16}.$$

The anti-Hermitian part of h_k is $-\gamma^4(k_5\gamma^5 + k_6\gamma^6 + k_7\gamma^7)$, whose square is $-(k_5^2 + k_6^2 + k_7^2)I_{16}$. Hence h_k is Hermitian, and commutes with B , if and only if $k_5 = k_6 = k_7 = 0$.

Result 10.5. These identities hold exactly for symbolic m and k (Wolfram) and by a structural proof for all real m, k plus 10 exact samples (Python); 2 of the samples with extra-time momentum have $E^2 < 0$.

checks

QNT_flatModeHamiltonian: wolfram-algebra, python-algebra

The good sector is the dimensional reduction to fields that do not depend on x_5, x_6, x_7 ($k_5 = k_6 = k_7 = 0$). It is not a subspace of the one-particle space $L^2(\mathbb{R}^7) \otimes \mathbb{C}^{16}$ of the seven-dimensional slices: $k_5 = k_6 = k_7 = 0$ is a set of measure zero there, so no normalizable state of the full theory lies in it, and a field independent of x_5, x_6, x_7 cannot satisfy the δ^7 anticommutator of Section 10.5. It is the reduced (4+1)-dimensional theory on the coordinates $(x_0, x_1, x_2, x_3, x_4)$; equivalently, one keeps only the zero modes in x_5, x_6, x_7 on a normalization volume V_3 , for which $\{\Psi_a, \chi_b\} = \delta_{ab} \delta^4/V_3$ (derived). There h_k is Hermitian and traceless with $h_k^2 = E_k^2$, $E_k = \sqrt{m^2 + k_0^2 + k_1^2 + k_2^2 + k_3^2}$, so its eigenvalues are $+E_k$ and $-E_k$ with multiplicity 8 each: 8 particle and 8 antiparticle states for every momentum $k = (k_0, k_1, k_2, k_3)$.

10.7 Fock space, Dirac sea and normal ordering

In the good sector (flat space, Gaussian normal time) let $u_s(k)$ and $v_s(k)$, $s = 0, \dots, 7$, be orthonormal (for $f^\dagger h$) eigenvectors of h_k with eigenvalues $+E_k$ and $-E_k$. The mode expansion

$$\Psi(x) = \int \frac{d^4 k}{(2\pi)^4} \sum_{s=0}^7 \left[b_s(k) u_s(k) e^{-iE_k x_4} + d_s(-k)^* v_s(k) e^{iE_k x_4} \right] e^{ik \cdot x},$$

$$\{b_s(k), b_{s'}(k')^*\} = \{d_s(k), d_{s'}(k')^*\} = \delta_{ss'} (2\pi)^4 \delta^4(k - k'),$$

with the Hilbert adjoints $*$ of the positive structure $J = B$, realizes the anticommutator of the reduced theory, $\{\Psi_a, \chi_b\} = \delta_{ab} \delta^4$ (Section 10.6, with the zero-mode normalization V_3 absorbed into the field), on a positive-definite Fock space. This Fock space is the quantization of the reduced (4+1)-dimensional theory, not a subspace of a Fock space of the full theory. The vacuum $|0\rangle$ is annihilated by all $b_s(k)$ and $d_s(k)$: the negative-energy states are filled (Dirac sea), and d^* creates antiparticles. After normal ordering

$$H = \int \frac{d^4 k}{(2\pi)^4} E_k \sum_{s=0}^7 (b_s(k)^* b_s(k) + d_s(k)^* d_s(k)) \geq 0.$$

This construction is derived here from Results 10.2, 10.4 and 10.5; it is not a separate machine check.

10.8 Unitary and Krein-unitary symmetries

A spin transformation that maps the slices $x_4 = \text{const}$ to themselves, $R = \exp(\theta S^{ab})$ with $a, b \neq 4$ acting on Ψ together with the corresponding transformation of the slice coordinates, preserves the canonical anticommutator iff $RBR^\dagger = B$, equivalently $R^\dagger BR = B$ (Krein-unitary). It is implemented unitarily on the positive Fock space iff in addition R is unitary for $f^\dagger h$ and commutes with $J = B$ (derived). These pointwise equal-time conditions apply only to such slice-preserving transformations. The boosts S^{a4} tilt the quantization slice, and for them the criterion is the one of ordinary Dirac theory, whose boost generators are Hermitian as well and whose boosts are

nevertheless unitarily implemented (see the end of this section). For $R = \exp(\theta S^{ab})$ the matrix conditions concern the generators.

Result 10.6.

1. Exactly 13 of the 28 generators S^{ab} commute with B : the 6 of $\mathfrak{so}(4)$ ($a, b \in \{0, 1, 2, 3\}$), the 3 of $\mathfrak{so}(3)$ ($a, b \in \{5, 6, 7\}$) and the 4 boosts S^{a4} , $a \in \{0, 1, 2, 3\}$. Since $B \propto \gamma^0 \gamma^1 \gamma^2 \gamma^3 \gamma^4$, the boosts S^{a4} anticommute with C and with γ^4 and therefore commute with B ; they are Hermitian, so the 16 by 16 matrix $\exp(\theta S^{a4})$ is neither unitary nor Krein-unitary. This is a statement about matrices, not about the physical implementability of the boosts.
2. Exactly 9 generators are anti-Hermitian (unitary for $\Psi^\dagger \Psi$) and commute with B . They generate $\text{Spin}(4) \times \text{Spin}(3)$, the compact slice-preserving part of the unitarily implemented symmetry group described below. The literal claim "exactly 9 of the 28 S^{ab} commute with B " is false and is recorded as false in both reports.
3. Exactly 21 generators, those with $a, b \neq 4$, satisfy the Krein condition $(S^{ab})^\dagger B + B S^{ab} = 0$. They generate the Krein-unitary slice group $\text{Spin}(4, 3)$. The Krein condition fails for all 7 generators S^{4b} .
4. Exactly 12 generators are anti-Hermitian for $\Psi^\dagger \Psi$ ($\mathfrak{so}(4)$ and $\mathfrak{so}(4)$ on the directions 4 to 7).

Both algebra reports use this meaning of the check, as recorded in stage1-summary.json. Time translations in the good sector are unitary (h_k Hermitian), and the $U(1)$ phase is unitary.

checks

QNT_unitaryAndKreinSubgroups: wolfram-algebra, python-algebra

Boosts in the good sector (derived; the matrix statements were checked numerically at sample momenta while revising this document, not as recorded checks). With $\beta := BC = -i\gamma^4$ and $\alpha^a := -\gamma^4 \gamma^a$ ($a = 0, \dots, 3$), all Hermitian, $\{\alpha^a, \alpha^b\} = 2\delta^{ab}$, $\{\alpha^a, \beta\} = 0$ and $\beta^2 = 1$, the good-sector mode Hamiltonian is $h_k = m\beta + \sum_{a=0}^3 k_a \alpha^a$: the standard Dirac Hamiltonian in 4+1 dimensions, on 16 components. A boost $R = \exp(\theta S^{a4})$ with $a \leq 3$ is not unitary, but $R^\dagger \beta R = \beta$. It maps the eigenspace of h_k with eigenvalue $\pm E_k$ onto the eigenspace of $h_{\Lambda k}$ with eigenvalue $\pm E_{\Lambda k}$, where Λk is the boosted momentum (it has no x_5, x_6, x_7 components), and on these eigenspaces $R^\dagger R = (E_{\Lambda k}/E_k) \cdot 1$. Hence $(U\psi)(\Lambda k) = (E_k/E_{\Lambda k}) R\psi(k)$ is unitary on $L^2(d^4 k)$ for each sign of the energy, and the boosts are unitarily implemented on the Fock space of Section 10.7 with an invariant vacuum, exactly as in ordinary Dirac theory. The symmetries of the eight-dimensional theory that act unitarily in the good sector therefore include $\text{ISO}(4, 1) \times \text{Spin}(3) \times U(1)$: the translations in $x_0, \dots, x_4$, the $\text{Spin}(4, 1)$ generated by the 10 S^{ab} with $a, b \in \{0, \dots, 4\}$, the $\text{Spin}(3)$ generated by the 3 S^{ab} with $a, b \in \{5, 6, 7\}$, and the phase. These 13 generators are exactly the ones of item 1 that commute with B ; $\text{Spin}(4) \times \text{Spin}(3)$ of item 2 is only their compact slice-preserving part. (The reduced theory has further internal symmetries that do not come from the S^{ab} : the commutant of $\gamma^0, \dots, \gamma^4$ has dimension 8.) The counts 13, 9, 21 and 12 of Result 10.6 remain correct as matrix facts.

10.9 The U(1) current and charge

$J^\mu = -i\bar{\Psi}\gamma^\mu\Psi$ is Hermitian, and in Gaussian normal gauge $J^4 = \Psi^\dagger B\Psi$: all eight current matrices are Hermitian and the matrix of J^4 is B . Hermitian means here, as in Sections 7.4 and 9.3, invariant under the conjugation of the Grassmann algebra, which after quantization is self-adjointness for the Krein form. In the positive Hilbert-space representation of Section 10.5 the operator $\Psi^\dagger M\Psi = \chi BM\Psi$ is self-adjoint only if BM is Hermitian, that is, for Hermitian M , only if $[M, B] = 0$. $J^0, \dots, J^4$ satisfy this; for J^5, J^6, J^7 the matrix $BM = -\gamma^4\gamma^\mu$ is anti-Hermitian, so these three components are anti-self-adjoint in the $J = B$ Fock space and are not observables there (Section 10.11). The conserved charge is $Q = \int d^7x \sqrt{|g|} J^4$. As a quadratic form in classical spinors it is indefinite (signature (8,8)); in the Hilbert representation of Section 10.5, $\Psi^\dagger B\Psi = \chi\Psi$, and after normal ordering $Q = \sum(b^*b - d^*d)$: particles have charge +1 and antiparticles -1 (derived).

checks

QNT_currentHermiticity: wolfram-algebra, python-algebra
GR_currentHermitian: grassmann-demo

10.10 The extra-time sector

For momenta along x_5, x_6, x_7 the operator h_k is not Hermitian and does not commute with B (Result 10.5), and $E^2 = m^2 + k_0^2 + \dots + k_3^2 - k_5^2 - k_6^2 - k_7^2$ can be negative: two of the Python samples have $E^2 = -271/196$ and $E^2 = -3$. Three cases occur, all with $h_k^2 = E^2 \cdot 1$ (Result 10.5). For $E^2 > 0$ the frequencies $\pm E$ are real although h_k is not Hermitian; h_k is diagonalizable because $x^2 - E^2$ has distinct roots (sample $m = 1$, $k_0 = 1/2$, $k_6 = -2/3$: $E^2 = 29/36$). For $E^2 = 0$ and $h_k \neq 0$, h_k is nilpotent, $h_k^2 = 0$, and the modes grow linearly, $\Psi(x_4) = (1 - ih_k x_4)\Psi(0)$ (samples $m = 1$, $k_5 = 1$ and $m = 0$, $k_1 = k_5 = 3$). For $E^2 < 0$, that is exactly when $k_5^2 + k_6^2 + k_7^2 > m^2 + k_0^2 + \dots + k_3^2$, the frequencies are imaginary and the modes grow exponentially in x_4 . This is the ill-posedness of the initial-value problem for ultrahyperbolic equations: the slice $x_4 = \text{const}$ contains three time-like directions. The positive Fock space of Section 10.7 exists only in the good sector.

10.11 Expectation values

In the Hilbert representation the canonical bilinear is $\Psi^\dagger M\Psi = \chi BM\Psi$. For a normalized one-particle state with wave function u ($u^\dagger u = 1$, positive energy, normal ordering),

$$\langle \Psi^\dagger M\Psi \rangle = u^\dagger B M u.$$

Examples (derived): the charge ($M = B$) gives $u^\dagger u = 1$; the energy ($M = Bh$) gives $u^\dagger h u = E$; the scalar density ($M = C$) gives $u^\dagger B C u = u^\dagger (-i\gamma^4)u$, which is +1 for a positive-energy state at rest with $m > 0$, so $\langle \rho \rangle = m\langle S \rangle = m = E$. The naive density $\Psi^\dagger \Psi$ gives $u^\dagger B u$, which is indefinite even on positive-energy states (Result 10.4); it is not the physical density.

For Hermitian M the rule gives real expectation values only when $[M, B] = 0$, because only then is BM Hermitian. Every $\Psi^\dagger M\Psi$ with M anticommuting with B is anti-self-adjoint in the $J = B$ Fock space and has purely imaginary expectation values, so it is not an observable there. Examples are J^5, J^6, J^7 (Section 10.9) and the T_{ij} bilinears of Section 9.7 with $i \in \{0, \dots, 3\}$ and $j \in \{5, 6, 7\}$.

For $(i, j) = (0, 5)$, $M = C\gamma^0\gamma^5\gamma^4$ is Hermitian, $BM = i\gamma^0\gamma^5$ is anti-Hermitian and commutes with h_0 , and its eigenvalues on the positive-energy rest space are $\pm i$ (derived; checked numerically while revising this document, not a recorded check).

11 Verification records

All five reports have schemaVersion 1, and every check in them is true. The counts below are those of stage1-summary.json, which re-reads the five reports, re-compares the key measurements and records an empty list of disagreements:

report	producer	checks true
wolfram-algebra-report.json	scripts/verify_dirac16complex_algebra.wls	21 of 21
wolfram-geometry-report.json	scripts/verify_dirac16complex_geometry.wls	43 of 43
python-algebra-report.json	scripts/check_dirac16complex_algebra.py	21 of 21
python-geometry-report.json	scripts/check_dirac16complex_geometry.py	52 of 52
grassmann-demo-report.json	scripts/demo_grassmann_lagrangians.py	16 of 16
stage1-summary.json	scripts/build_stage1_summary.py	153 of 153

The producers record Wolfram Language 15.0.1 and Python 3.14.5, sympy 1.14.0, numpy 2.4.6.

11.1 WolframScript, algebra

The Wolfram algebra verifier uses exact integer, rational and Gaussian-rational arithmetic. Commutants, intertwiners and invariant forms are exact null spaces of integer coefficient matrices, and signatures come from exact characteristic polynomials. Files and checks:

```
wolfram/Dirac16ComplexAlgebra.wl
scripts/verify_dirac16complex_algebra.wls
```

```
wolfram-algebra: ALG_clifford ALG_gammaTransposeSymmetry ALG_chargeMatrix ALG_expression1
  ALG_spinTransposeProperties ALG_chirality ALG_faithful ALG_pinIrreducibleComplex
  ALG_spinDecomposition ALG_cliffordPictureIntertwiner ALG_octonionPictureIntertwiner
  ALG_chargeFormB ALG_invariantForms ALG_gamma8Map ALG_pinLiftCharacter
  QNT_curvedAnticommutatorMatrix QNT_flatModeHamiltonian QNT_kreinSignature
  QNT_unitaryAndKreinSubgroups QNT_currentHermiticity ALG_fixtureAgreement
```

11.2 WolframScript, geometry

Every derived object of the Wolfram geometry verifier is an exact order-1 jet (value and eight first derivatives) at the test point, and the G2 arithmetic is exact in $\mathbb{Q}(\cos z)$. The Euler-Lagrange, canonical-momentum and Lg[] checks use a genuine Grassmann algebra; the energy-momentum, Lichnerowicz and invariance checks use commuting exact numbers, which is exact for bilinears with $\Psi^\dagger$ on the left and Ψ on the right (measurement convention.commutingEvaluation). Files and checks:

```
wolfram/Dirac16ComplexGeometry.wl
scripts/verify_dirac16complex_geometry.wls
```

```
wolfram-geometry: ALG_chargeMatrix ALG_cliffordRelations ALG_grassmannLemmas
  EMT_conservation_G1 EMT_conservation_G2 EMT_homogeneousReduction_G3
  EMT_symmetricHermitian_G1 EMT_symmetricHermitian_G2 EMT_trace_G1 EMT_trace_G2
  EMT_variation_G3 GEO_curvature_G1 GEO_curvature_G2 GEO_diagonalSlashFormula_G2
  GEO_divergenceIdentity_G1 GEO_divergenceIdentity_G2 GEO_frameNondegenerate_G1
  GEO_frameNondegenerate_G2 GEO_gammaCovariantConstancy_G1
  GEO_gammaCovariantConstancy_G2 GEO_lichnerowicz_G1 GEO_lichnerowicz_G2
  GEO_notebookContractionFails_G1 GEO_notebookContractionFails_G2
  GEO_omegaAntisymmetry_G1 GEO_omegaAntisymmetry_G2 GEO_primordialInvariants_G2
  GEO_symbolicJetAgreement_G2 GEO_vielbeinPostulate_G1 GEO_vielbeinPostulate_G2
  LAG_eulerLagrangePsibar_G1 LAG_eulerLagrangePsibar_G2 LAG_eulerLagrangePsi_G1
  LAG_eulerLagrangePsi_G2 LAG_hermiticity_G1 LAG_hermiticity_G2
  LAG_localSpinInvariance_G1 LAG_localSpinInvariance_G2
  LAG_notebookLgGrassmannTrivial_G1 LAG_notebookLgGrassmannTrivial_G2
  NEG_notebookConnectionDetected_G2 QNT_canonicalMomentum_G1 QNT_canonicalMomentum_G2
```

11.3 Python, algebra

The Python algebra checker uses the standard library only (exact rationals, no floating point). Files and checks:

```
scripts/d16c_exact.py
scripts/build_dirac16complex_fixture.py
scripts/check_dirac16complex_algebra.py
```

```
python-algebra: ALG_clifford ALG_gammaTransposeSymmetry ALG_chargeMatrix ALG_expression1
  ALG_spinTransposeProperties ALG_chirality ALG_faithful ALG_pinIrreducibleComplex
  ALG_spinDecomposition ALG_cliffordPictureIntertwiner ALG_octonionPictureIntertwiner
  ALG_chargeFormB ALG_invariantForms ALG_gamma8Map ALG_pinLiftCharacter
  QNT_flatModeHamiltonian QNT_kreinSignature QNT_unitaryAndKreinSubgroups
  QNT_currentHermiticity ALG_fixtureAgreement ALG_wolframAgreement
```

11.4 Python, geometry

The Python geometry checker uses sympy with exact rationals: G1 by exact truncated Taylor jets, and G2 at a fully symbolic point in $\mathbb{Q}(w, c, E, A_1, A_2, A_3, H)$ with $w = \sin^{1/6} z$, $c = \cos z$, $E = e^{a_4}$ and A_k the derivatives of a_4 . The relation $w^{12} + c^2 = 1$ was needed in 0 of its zero tests. Files and checks:

```
scripts/d16c_geometry_sympy.py
scripts/check_dirac16complex_geometry.py
```

```
python-geometry: ALG_cliffordRelations ALG_sigma16EqualsGamma0123
  ALG_CSymmetricInvolution ALG_expression1CgammaAntisymmetric ALG_gammaSymmetryPattern
  ALG_CSabAntisymmetric ALG_chiralityDiag ALG_CAnticommutatorGammaSSymmetric
```

```

ALG_CCommutatorGammaSAntisymmetric ALG_SabGammaCommutator GEO_frameNondegenerate_G1
GEO_vielbeinPostulate_G1 GEO_omegaAntisymmetry_G1 GEO_gammaCovariantConstancy_G1
GEO_notebookContractionFails_G1 GEO_divergenceIdentity_G1
GEO_sqrtgSquaredEqualsDetg_G1 GEO_curvature_G1 GEO_lichnerowicz_G1
LAG_eulerLagrangePsibar_G1 LAG_eulerLagrangePsi_G1 EMT_onshellJetSolve_G1
EMT_symmetric_G1 EMT_trace_G1 EMT_traceOffShellIdentity_G1
EMT_onshellLagrangianSUprimeMinusU_G1 EMT_conservation_G1 GEO_vielbeinPostulate_G2
GEO_omegaAntisymmetry_G2 GEO_gammaCovariantConstancy_G2
GEO_notebookContractionFails_G2 GEO_divergenceIdentity_G2
GEO_sqrtgSquaredEqualsDetg_G2 GEO_curvature_G2
GEO_anticommutatorGammaOmegaVanishesDiagonal_G2 GEO_primordialInvariants_G2
GEO_diagonalSlashFormula_G2 GEO_lichnerowicz_G2 LAG_eulerLagrangePsibar_G2
LAG_eulerLagrangePsi_G2 EMT_onshellJetSolve_G2 EMT_symmetric_G2 EMT_trace_G2
EMT_traceOffShellIdentity_G2 EMT_onshellLagrangianSUprimeMinusU_G2
EMT_conservation_G2 GEO_lichnerowiczConstantSameG1G2 EMT_homogeneousReduction
GEO_diagonalSlashFormula_G3 EMT_homogeneousTimeSpaceVanishesOnShell_G3 EMT_variation
GEO_wolframAgreement

```

11.5 Python, Grassmann demonstrations

The demonstrations work in exact Grassmann algebras with 720 generators for real Ψ jets and 1440 for complex (Ψ, Ψ^*) jets. Files and checks:

```

scripts/grassmann_algebra.py
scripts/demo_grassmann_lagrangians.py

```

```

grassmann-demo: GR_conjugationRules GR_massTermVanishesReal
GR_bilinearOnlyAntisymmetricPartSurvives GR_kineticTotalDerivativeReal
GR_kineticSymmetricMatrixContrast GR_quarticTermPolynomial GR_scalarBilinearHermitian
GR_currentHermitian GR_notebookLgELTrivial GR_notebookLgPureDivergence
GR_complexLagrangianNonTrivial GR_complexQuarticEL GR_complexPsiEquation
GR_lagrangianHermitian GR_emtHermitian GR_unsymmetrizedKineticNotHermitian

```

11.6 Agreement between the implementations

The cross-implementation checks:

```

ALG_fixtureAgreement (wolfram-algebra):  Wolfram reads the Python fixture;
75 of 75 matrices agree by name
ALG_wolframAgreement (python-algebra):  Python reads the Wolfram algebra report;
K_clifford, K_octonion and the chirality agree; 20 shared check names
with equal verdicts; 47 of 47 shared measurements agree
GEO_wolframAgreement (python-geometry):  Python reads the Wolfram geometry report;
26 shared check names with equal verdicts;
84 of 84 shared measurements agree

```

The geometry comparison covers the G1 point data, the nonzero counts, the Lichnerowicz constant, the curvature sign, the Python G2 formulas evaluated at the exact Wolfram points, and the energy-momentum sign and trace conventions. The field-dependent values (traces, ρ , p , KE and PE at the

test points) are not compared between the implementations, because each draws its own random field data and parameters; they are compared through the formulas that both verify. stage1-summary.json records 30 agreed key measurements, the hash links between the reports, and the equivalences that hold after a notation conversion.

11.7 Files and hashes

sha256 of the reports, of the fixture and of the producers, as recorded in the reports and in stage1-summary.json:

```
artifacts/dirac16complex/arbitrary-field/wolfram-algebra-report.json
d43adeeba580bdd8cec56fac5fa74906b7578c06c8589e52dbe3da1404cd49e7
artifacts/dirac16complex/arbitrary-field/wolfram-geometry-report.json
cec9ee0d577c4e7f0e82efd404d412a503140ccb1547118ab48a78a3d773f0ac
artifacts/dirac16complex/arbitrary-field/python-algebra-report.json
3275b2af1a2f56f9bd779ad359e7ed078cb2c64c744863732f789f410bb50c97
artifacts/dirac16complex/arbitrary-field/python-geometry-report.json
deec576c1613bb06ce8ac4ad9f89a7b008239bd9d4b624bea14728fa77b2a197
artifacts/dirac16complex/arbitrary-field/grassmann-demo-report.json
b83997857ccfc9fb71f05efbcaebe16f29bf36687cc08c16e0823a26a6369f09
artifacts/dirac16complex/arbitrary-field/stage1-summary.json
a65972b5a52f947d9bee9f67e09f7151fff617c03f496ba313e07db168c10770
artifacts/dirac16complex/arbitrary-field/algebra-fixture.json
8b4f15462ca4d61ef6ec0e72f04c8a77d23ce8bcabc9b6ce19f921c90e02653b
wolfram/Dirac16ComplexAlgebra.wl
ba0d00c818fa3707f04f7b2a7478b0c635fbf9fdaaa08b2e036d2704180f2262
scripts/verify_dirac16complex_algebra.wls
86904ac07e1237cf0dec14431a3707d7bf24f951e12def504705d331c934949a
wolfram/Dirac16ComplexGeometry.wl
f5b674665eee4000750161e6ab6312c38bfac9da7a17450a2b3bdd88ef292af2
scripts/verify_dirac16complex_geometry.wls
81b1b0febf1a5d6956f042179dff4eed949718cc15955e0753c71606f3f09953
scripts/d16c_exact.py
eb2fcbe498b6ae927df61f73abcee4e58c319984fb8f6e1b28defcd0ca13a3f5
scripts/build_dirac16complex_fixture.py
585051a6b2e01ab1d8ae92ae5e1949c4d0c23e37815332c44dc5af525a36d776
scripts/check_dirac16complex_algebra.py
3d2fd639b655491897b68f8719db3618d638248b2b1176f22668723e273bea12
scripts/d16c_geometry_sympy.py
4f7eca4cf247354650ed9beb0b9c46530d14c9965f095979ea5ee24651af9f25
scripts/check_dirac16complex_geometry.py
ac26111ce6758cebdf473e943d1cd4f1e14f87e4961ee694623d888742824653
scripts/grassmann_algebra.py
dfce851d3dc7ab82202a62d414cc259644a64a101f08e9ba80ed747faa8bc24c
scripts/demo_grassmann_lagrangians.py
3053c7011037e29cd9efa7c2067aafd23ebc67c6156eb581fef184810ef1515a
dirac-main/artifacts/exact/cl44-seed.json
52e7a73b14467435d7986b33bf29547510b321853ffbd76121896d70b796d133
dirac-main/artifacts/exact/split-octonion.json
cbc28db6709ec1fc35ccfcc26353d1952b42bb516f6c50d6a52e2d6d355be710
dirac-main/artifacts/exact/triality44.json
a8c31c07809fc884ae0495d91f4105fac0909c954bb185c643306c57699cd395
```

12 Reproduction

Run every command from the repository root.

Prerequisite: `dirac-main`. The algebra verifiers compare with three files of `dirac-main`, the separately published repository <https://github.com/once-ere/dirac>. It is git-ignored here (`/dirac-main/` in `.gitignore`), so a fresh clone does not contain it: clone that repository into `./dirac-main`, or unpack the GitHub archive of its main branch there. The recorded run did not record a commit of `dirac-main`; the pin is the content of the three files, whose sha256 must equal the values in Section 11.7 (`dirac-main/artifacts/exact/cl44-seed.json`, `split-octonion.json` and `triality44.json`; check with `sha256sum` in Git Bash or `Get-FileHash` in PowerShell). Without `dirac-main` every verifier still exits 0, but the `dirac-main` sub-checks record "not-run", both algebra reports differ in their bytes from the recorded ones (so their sha256 differ from Section 11.7), the `dirac-main` equalities of Results 3.7 and 3.8 are not reproduced, and the gate fails at step 08, whose publication test compares the reports with this document, and prints `stage1_arbitrary_field_verification=FAILED`.

The Wolfram report path is a plain positional argument: WolframScript 1.14 drops "-" and every argument after it when it is combined with `-file`. The gates set `PYTHONUTF8=1` for their own process; set it yourself for single steps, because Python output contains Greek letters.

12.1 PowerShell

The Stage-1 gate runs all twelve steps through the logging wrapper scripts/`run_logged.ps1`, with the logs in `build/logs/`, and then audits the five reports and the cross-implementation checks. Step 12 repeats the publication test of this document after the summary (step 10) and the PDF (step 11) have been rebuilt, so that the hash of `stage1-summary.json` quoted in Section 11.7 is compared with the file the gate has just written:

```
pwsh -NoProfile -File scripts/verify_stage1_arbitrary_field.ps1
```

Its last line is

```
stage1_arbitrary_field_verification=OK
```

The individual steps:

```
$env:PYTHONUTF8 = "1"
$a = "artifacts/dirac16complex/arbitrary-field"
python scripts/build_dirac16complex_fixture.py
python scripts/check_dirac16complex_algebra.py --wolfram-report=
wolframscript -file scripts/verify_dirac16complex_algebra.wls '
    "$a/wolfram-algebra-report.json"
wolframscript -file scripts/verify_dirac16complex_geometry.wls '
    "$a/wolfram-geometry-report.json"
python scripts/check_dirac16complex_geometry.py --wolfram-report '
    "$a/wolfram-geometry-report.json"
python scripts/demo_grassmann_lagrangians.py
python scripts/check_dirac16complex_algebra.py --wolfram-report '
```

```

"$a/wolfram-algebra-report.json"
python -m unittest discover -s tests -p "test_d16c_[ag]*.py" -v
python -m unittest discover -s tests -p "test_publication_tooling.py" -v
python scripts/build_stage1_summary.py --output "$a/stage1-summary.json"
python scripts/build_provenance_pdf.py provenance/DIRAC16COMPLEX_ARBITRARY_FIELD.md
python -m unittest discover -s tests -p "test_d16c_arbitrary_field_publication.py" -v

```

12.2 Git Bash

The Git Bash twin runs the same steps through scripts/run_logged.sh and ends with the same line:

```
bash scripts/verify_stage1_arbitrary_field.sh
```

The individual steps:

```

export PYTHONUTF8=1
a=artifacts/dirac16complex/arbitrary-field
python scripts/build_dirac16complex_fixture.py
python scripts/check_dirac16complex_algebra.py --wolfram-report=
wolframscript -file scripts/verify_dirac16complex_algebra.wls \
    "$a/wolfram-algebra-report.json"
wolframscript -file scripts/verify_dirac16complex_geometry.wls \
    "$a/wolfram-geometry-report.json"
python scripts/check_dirac16complex_geometry.py --wolfram-report \
    "$a/wolfram-geometry-report.json"
python scripts/demo_grassmann_lagrangians.py
python scripts/check_dirac16complex_algebra.py --wolfram-report \
    "$a/wolfram-algebra-report.json"
python -m unittest discover -s tests -p "test_d16c_[ag]*.py" -v
python -m unittest discover -s tests -p "test_publication_tooling.py" -v
python scripts/build_stage1_summary.py --output "$a/stage1-summary.json"
python scripts/build_provenance_pdf.py provenance/DIRAC16COMPLEX_ARBITRARY_FIELD.md
python -m unittest discover -s tests -p "test_d16c_arbitrary_field_publication.py" -v

```

Each verifier prints `check_count` and `failed_check_count` and exits nonzero if a check fails. The first algebra step runs without the Wolfram comparison (`--wolfram-report=` with an empty value), because the Wolfram report on disk may predate the run; the second one compares with the report just written. The PDF step builds this document twice (two builder runs, three `pdflatex` passes each), requires warning-free logs and byte-identical PDFs, and compares the result with the registered edition `dirac16complex-arbitrary-field` in `provenance/pdf-specifications.json`. After an edit of this document the edition is registered again with

```
python scripts/build_provenance_pdf.py --register \
    provenance/DIRAC16COMPLEX_ARBITRARY_FIELD.md
```

(in PowerShell the line continuation is a backtick instead of the backslash), and the sha256 pins in `tests/test_d16c_arbitrary_field_publication.py` are updated.

13 Limitations

1. The general statements of Sections 5 to 9 are proved by the arguments given here and checked exactly at test points: three points of one generic non-diagonal vielbein (G1), the primordial family (G2, symbolic in Python) and three times of a diagonal frame (G3). The checks are not a symbolic verification for every vielbein.
2. The metric variation of the action is carried out explicitly only for diagonal metrics. The off-diagonal components of $T_{\mu\nu}$ are tested through symmetry, Hermiticity and on-shell conservation in G1, and through the homogeneous forms of T_{ij} and T_{4i} .
3. Conservation of $T_{\mu\nu}$ is checked with on-shell jets at points, not as an off-shell Noether identity.
4. The energy-momentum, Lichnerowicz and invariance checks evaluate bilinears with commuting exact numbers, which is exact because $\Psi^\dagger$ is always on the left and Ψ on the right; the field equations, the canonical momentum and $\text{Lg}[]$ are checked in genuine Grassmann algebras.
5. The Dirac-bracket computation, the Hamiltonian density, the Heisenberg equations, the mode expansion, the Fock space and the expectation-value rule of Section 10 are derivations; only their algebraic ingredients (Results 10.1 to 10.6) are machine-checked.
6. The quantization is formal: the state space is a Krein space; in the extra-time sector the mode Hamiltonian is not Hermitian, with imaginary frequencies (exponential growth) when $k_5^2 + k_6^2 + k_7^2 > m^2 + k_0^2 + \dots + k_3^2$ and linear growth at equality; the positive Fock space is that of the reduced good sector; and no interacting theory, regularization or renormalization is constructed.
7. The homogeneous-sector formulas treat bilinears as classical mean-field quantities; the quantum expectation values of Section 10.11 are one-particle statements.
8. The signature (4,4) is not observed spacetime, and nothing here is compared with data. The connection to dark energy or dark matter, and the stability of solutions, are not decided in this document.
9. The cross-checks against dirac-main's published fixtures need the git-ignored dirac-main folder, which Section 12 says how to obtain. Without it these sub-checks record "not-run" (measurement referenceFilesPresent in wolfram-algebra-report.json), the two algebra reports differ in their bytes and hashes from the recorded ones, the dirac-main equalities of Results 3.7 and 3.8 are not reproduced, and the gate fails at step 08 instead of printing OK.